From Brain Scores to Training Signals: A Controlled Test of Neural-Guided Distillation

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Abstract

Model representations of an language model (LM) can predict human brain responses to language, a measurable correspondence called brain alignment. Observing this alignment does not show that brain responses can improve model training. This thesis tests whether brain responses can improve LM compression by guiding a distilled student toward representations that remain effective for language and become more predictive of brain responses. We test two distillation approaches: one using recorded functional magnetic resonance imaging (fMRI) responses and another using a separate model to generate synthetic brain responses from text. We confirmed that middle-layer activations from pretrained LMs help predict fMRI responses to new sentences, even after accounting for simpler linguistic features of those sentences. After ordinary distillation, the student's middle-layer activations were less predictive of brain responses than the teacher's. Because the student was also a worse LM, the source of this gap remained unclear. Training with recorded responses did not reliably improve the student's ability to predict brain responses across participants. Training with synthetic brain responses changed the student's model representations while keeping language quality within our prespecified tolerance, and made the synthetic responses easier to recover from those representations. Yet a separate post-hoc linear-readout retention test did not establish that the synthetic-target student retained more information about the synthetic responses than a student trained by ordinary knowledge distillation (KD). Because the alternative target differed too much from the synthetic brain-response target, we could not determine whether the changes were specific to synthetic brain responses. Direct improvement over ordinary distillation on independently recorded fMRI responses was near zero across nine participants. Thus, the tested methods did not demonstrate a reliable, brain-specific training benefit across participants, but they do not show that brain-guided training is universally ineffective.

1 Introduction

Brain responses to language could help train a smaller language model (LM) by selecting among model representations that are similarly effective for predicting text. In knowledge distillation (KD), a smaller student learns to reproduce the output distribution of a larger teacher, but comparable output behavior does not require the two models to organize information identically in their middle layers [1]. Under learning using privileged information (LUPI), a recorded functional magnetic resonance imaging (fMRI) response or synthetic brain-response generated from text can serve as a training-only target: it may favor one language-compatible model representation over another without changing the inputs available at deployment. Feature-distillation and LUPI research make this possibility credible, while also showing that an auxiliary target can help for reasons unrelated to whether it contains information about brain responses [2, 3]. Brain-response supervision is thus best treated as a candidate representational regularizer, not as evidence that the student has acquired a human cognitive mechanism.

Brain alignment is assessed by fitting an encoding model that predicts recorded fMRI responses from frozen LM representations, then evaluating its predictions on stimuli withheld from fitting. Successful prediction indicates that the representations contain information useful for predicting recorded fMRI responses to the same stimuli [4, 5]. The resulting score characterizes the complete assay rather than the LM alone: it depends on the participants and response aggregation, stimulus distribution, data split, nuisance features, response reliability, readout class, and inference unit. In this manuscript, *controlled brain predictivity* refers to brain alignment after the specified nuisance, split, and control-model checks. Section 3.2 defines the main held-out scores and summarizes the readout protocol; Appendix B specifies the nuisance features and preprocessing, and Appendix D formalizes the estimators and inference.

A positive brain-alignment score shows that model representations contain information useful for predicting recorded fMRI responses under the specified assay. It does not establish that training with brain responses can alter the student's retained representations. Even if such a change occurs, the score does not establish three further claims: that the change is caused specifically by brain-response content rather than generic target geometry or optimization pressure; that it transfers to independently recorded fMRI responses; or that it improves language performance. These distinctions matter because inadequate data splits, simple stimulus features, and analysis choices can produce or reorder apparent model–brain correspondences [6, 7]. An encoding model should therefore be treated as a measurement instrument between model representations and recorded fMRI responses, not as a mechanism presumed to exist in either the LM or the brain.

Prior work does not resolve whether brain responses can help train a smaller model. Encoding studies show that model representations can predict recorded brain responses and examine which linguistic properties and model layers support that prediction [4, 5]. Brain-response-guided optimization studies show that recorded brain responses or synthetic brain responses generated from text can alter model representations or task behavior in several language and speech settings [8–11]. Research on privileged information and feature distillation shows more generally how dense training-only targets may benefit a student. Compression studies also show that brain alignment after pruning or quantization can depend on the compression method and model scale [2, 3, 12]. However, these studies differ in their modality, temporal resolution, participant specificity, model budget, endpoint, comparator, training seeds, and inference unit. They therefore do not establish whether training with brain responses gives a fixed smaller student an incremental advantage over a matched text-derived control at comparable LM quality.

We therefore evaluate a brain-specific training advantage as an ordered chain of evidence. First, the model representations must show controlled brain predictivity, and the smaller model must leave room for improvement that cannot be explained only by worse LM quality. For synthetic brain responses generated from text, we then ask whether the exact training target contains measurable information about recorded fMRI responses, whether the student learns that target, and whether training produces changes that remain after the temporary prediction head is removed. Finally, training with synthetic brain responses must be compared with training that uses a similarly constructed text-derived auxiliary target. Because both conditions retain the same LM objective, this comparison tests whether any difference is specific to brain-response content. Tests must then determine whether those changes transfer to independently recorded fMRI responses and improve an external task. Failure at an earlier stage limits what evidence from later stages can establish.

In the tested systems, trained LM representations contained information useful for predicting held-out recorded fMRI responses beyond the specified nuisance variables and untrained-model controls. The distilled student was less brain-predictive than its teacher, but worse LM quality remained a competing explanation for this gap. Training with recorded fMRI responses did not produce a reliable participant-level benefit in the tested regimes. For synthetic brain responses generated from text, the exact training target did not meet the prespecified threshold for additional prediction of recorded fMRI responses beyond nuisance variables under the specified linear assay. The target was nevertheless learnable on the training corpus, and training changed the student's retained representations while preserving LM quality within the prespecified tolerance. A fresh post-training readout on the Tuckute data did not resolve whether the student retained more information about the target than a student trained by ordinary KD. The available text-derived auxiliary target was not sufficiently comparable to determine whether the observed differences arose specifically from brain-response content. The direct participant-level comparison between training with synthetic brain responses and ordinary KD found a near-zero difference in prediction of recorded fMRI responses.

This study contributes controlled measurement of brain predictivity alongside LM quality, participant-level tests of training with recorded fMRI responses, and an evaluation framework that locates where evidence for training with synthetic brain responses is supported, unresolved, or absent. The conclusion is restricted to the tested models, English stimuli, datasets, participants, targets, objectives, optimization regimes, controls, linear readouts, comparators, inference units, and endpoints. It does not provide a successful brain-response-guided distillation method, but neither does it show that brain responses can never help train a smaller model.

2 Evidence Required for a Brain-Specific Training Advantage

Brain-encoding [4, 5], brain-response-guided optimization [8–11], privileged-information and feature-distillation [2, 3], and model-compression [12] studies address different questions. Predicting recorded brain responses does not establish that training with those responses changes a student model. A change in the student also does not establish that it arises specifically from brain-response content, transfers to independently recorded fMRI responses, or improves a useful endpoint at comparable LM quality. This section connects these claims and defines the evidence required to support a brain-specific training advantage.

2.1 What controlled brain predictivity establishes

Controlled brain predictivity tests whether representations extracted from a frozen LM contain information useful for predicting recorded fMRI responses to held-out stimuli. An encoding model is fitted to predict responses from those representations on one set of stimuli and evaluated on a separate set. This held-out evaluation tests whether prediction generalizes beyond the fitting stimuli rather than merely reflecting overfitting to them [4, 5]. Under a linear readout, a positive controlled score means that a linear combination of LM representation dimensions improves prediction beyond the declared controls.

Three controls are required to determine whether predictive performance is associated with learned LM representations rather than simpler alternatives. First, a nuisance-only model predicts recorded fMRI responses from stimulus properties such as position, length, and static lexical features. Comparing it with a full model that also includes the tested representations measures what those representations add. Second, an architecture-matched untrained LM retains the tokenizer, dimensionality, and architecture of the trained model but lacks its learned parameters. This comparison tests whether predictive performance depends on learned representations rather than the model's fixed structure. Third, contiguous or blocked splits keep locally related samples from being scattered across fitting and evaluation sets, reducing inflated scores caused by dependence or leakage between them [6]. Together, these controls narrow the observed effect to held-out predictive value added by learned LM representations under the declared assay.

Even with these controls, controlled brain predictivity remains bounded by the declared assay. A linear readout tests whether recorded responses can be predicted from a linear combination of LM representation dimensions; it does not test every form of information that might be present. Participant-specific responses measure predictivity for an individual, whereas averaged responses measure predictivity of a group-level response summary. The stimulus distribution, response definition and aggregation, response reliability, and selected brain regions further determine what can be predicted [13]. The inference unit determines the scope of uncertainty: many voxels or repeated folds from one participant do not provide evidence about many independent participants. Controlled brain predictivity therefore supports a bounded claim about held-out prediction under the specified assay, not a shared biological mechanism, causal equivalence, or general cognitive competence [7]. Sections 3.1, 3.2, and 3.5 specify the response construction, estimator, controls, and inference unit used to instantiate this assay.

2.2 Why brain-response targets might help a student

Ordinary KD trains a student to reproduce the teacher's output distribution, but this objective does not uniquely determine how the student organizes information in its middle layers. Different bases, subspaces, and feature organizations can support similar token predictions. Two students can therefore show similar language behavior while retaining different information in their model representations. This leaves room for an auxiliary training target to favor one representation over another without necessarily improving or degrading the language objective. Feature-distillation research further shows that the value of such guidance depends on which information is emphasized and how teacher and student capacities are matched, rather than simply on copying a large dense vector [3].

Brain-response targets can exploit this flexibility as privileged information available during training but unnecessary at deployment [2]. The student retains the same LM objective, while an auxiliary objective favors representations from which the training target can be predicted. Recorded fMRI responses provide observed biological measurements, whereas synthetic brain responses generated from text package structure learned by a fixed generator. Because each synthetic response is determined by the text and generator, it cannot provide example-specific information that varies independently of them. It may nevertheless present generator-learned structure in a form that is easier for the student to learn or recover; this is an intervention hypothesis motivated by privileged-information and feature-distillation results, not a consequence of the target's deterministic construction [2, 3].

Evidence that the auxiliary objective affected training does not establish a brain-specific benefit. Reduced target loss may show only that the model learned to predict the auxiliary target during training, without demonstrating a lasting change in the student. Target uptake instead requires a new prediction model fitted after training to recover the target from frozen retained-student representations relative to the declared baseline. Retained-student movement separately requires parameter or representation differences from ordinary KD that remain after temporary training components are removed. These checks show, respectively, that target information became recoverable beyond the baseline and that the intervention changed the retained student beyond ordinary KD. They do not establish that the changes arose specifically from brain-response content, transferred to independently recorded fMRI responses, or improved a practical endpoint. Sections 3.1, 3.3, 3.4, and 3.5 specify the target construction, training objectives, gradient paths, retained components, and comparison baselines used for these checks.

2.3 What prior brain-response-guided training leaves unresolved

Prior work assigns brain responses two distinct roles: post-training evaluation and training supervision. Post-training encoding studies use recorded fMRI responses to evaluate model representations after the model has been changed for another reason. Narrative-summarization tuning can increase brain-predictivity scores without using brain responses during training, while model scaling can increase naturalistic brain alignment even when matched-size instruction tuning does not under the studied assay [14, 15]. These findings show that brain predictivity can change because of the training objective or model scale alone; they do not estimate the value of brain responses as training targets. Brain-response-guided studies instead use recorded brain responses or model-generated brain-response predictions during training [8–11]. Their positive findings show that these objectives can change model representations, brain-predictivity scores, or selected task outcomes. However, the claim supported by each result depends on its comparator, evaluation data, and inference unit. Table 1 compares those designs without treating every positive brain-predictivity result as evidence for the same claim.

Table 1: Prior designs linking model interventions to brain predictivity.

Study	Intervention and relevant result	Why it does not settle this study's question
Aw and Toneva [14]; Gao et al. [15]	Post-training evaluation after summarization tuning, scaling, or instruction tuning shows that brain predictivity can change without brain-response training.	No brain-response target, smaller student, or brain-response-versus-control intervention.
Bilgin et al. [16]	Cosine loss against participant-specific fMRI responses and token loss update upper-layer adapters; held-out encoding, new-participant readouts, and a model-specific external task improve.	No shuffled or matched text-derived target, smaller student, or ordinary-KD comparator.
Merlin, Moussa, and Toneva [10]	Participant-specific fMRI-response, stimulus-language, or joint objectives update adapters; brain-response training wins more linguistic-probe comparisons than stimulus tuning.	Target geometry and difficulty are unmatched; no participant-held-out transfer or compression comparison.
Merlin and Toneva [17]	Complementary objectives add or remove brain-predictive information; a permuted-fMRI control and matched LM loss accompany changes in linguistic probes.	No smaller student; the manipulated readout may contain linguistic variance not specific to brain-response content.
Freteault et al. [18]	Individual or group fMRI responses fine-tune selected auditory-network layers; held-out brain predictivity and external auditory tasks improve.	No matched text-derived or shuffled-target arm; the model and modality differ from text distillation.
Pirlot et al. [19]	A monkey-V1-response regularizer trains a vision model; shuffled targets reproduce the clean-accuracy gain, while real targets improve moderate-attack robustness.	The clean-task gain is generic regularization; only the robustness endpoint distinguishes brain-response content in that design.

Positive studies span language, speech, and vision and do not estimate a common intervention effect. Speech models adapted with recorded fMRI responses have improved brain-response prediction and selected semantic or auditory tasks, including transfer across datasets or participants [8, 18, 20]. Text-model interventions report changes in multilingual performance, linguistic probes, visual commonsense, or reasoning after training with brain-response targets [9, 10, 16, 21]. Training with temporally precise electrocorticography responses has improved prediction within a speech substrate, while training with slower fMRI responses has transferred to prediction in that faster modality for new participants and stimuli after fitting the required readout [11, 22]. Together, these studies show that brain-response-guided objectives can alter model representations or selected task outcomes. However, they test different models, training targets, endpoints, and transfer settings. Some adapt a full encoder or adapters rather than a smaller student, some use participant-specific targets, and some require different amounts of target-participant data. Their positive results therefore do not establish one common effect or the incremental value of brain responses for a fixed smaller student.

Different controls rule out different explanations for a positive result. A target permutation destroys the stimulus–target pairing while preserving the target values and their marginal distribution, testing whether the correct pairing matters. A task-only or stimulus-language condition tests whether additional training or a conventional objective is sufficient. Evaluation on a held-out participant or dataset tests whether the change transfers beyond the training setting, often after fitting a new readout. None of these controls replaces a text-derived auxiliary target matched to the brain-response target in dimensionality, covariance, baseline predictability, learnability, and optimization pressure. The vision result is instructive: shuffled targets reproduce the clean-accuracy improvement, while only the robustness endpoint distinguishes the real brain-response target [19]. Positive results should therefore be retained within the claims supported by their designs, while attribution to brain-response content must be limited to the alternative explanations that their controls rule out.

Privileged-information and feature-distillation studies provide a competing explanation for any benefit from training with brain responses. A dense training-only target can help by emphasizing distinctions useful to the student or by making optimization easier, regardless of whether the target contains brain-response content. Its value depends on the information carried by the target, the student's capacity, and how the teacher and student representations are matched [2, 3]. An improvement from a brain-response target therefore requires comparison with a similarly learnable text-derived auxiliary target under the same LM objective before it can be attributed specifically to brain-response content.

Measurement-validity research supplies separate requirements for evaluating the student after training. Post-training brain predictivity must be measured with leakage-resistant splits, nuisance controls, and an inference unit appropriate to the claim [6, 7, 13]. Reduced auxiliary loss during training cannot replace this fresh evaluation. Compression studies further show that pruning or quantization method and model scale can change brain predictivity without brain-response training [12]. An intervention comparison must therefore separate the effect of the brain-response objective from changes in LM quality and must evaluate the retained student rather than reuse its training objective.

The unresolved pre-results question is therefore narrower than whether training with brain responses can change a model. For a fixed smaller student trained by ordinary KD, do recorded fMRI responses or synthetic brain responses generated from text provide an incremental, brain-specific advantage over the appropriate permuted or text-derived control at comparable LM quality? The next subsection states the evidence required to support that claim.

2.4 Evidence standard for a brain-specific training advantage

Figure 1 separates four roles in the evidence chain: opportunity prerequisites, manipulation checks, brain-response-specific attribution, and endpoint evidence. In the figure, stages 1–3 are opportunity prerequisites, stages 4–5 are manipulation checks, stage 6 tests brain-response-specific attribution, and stages 7–8 test biological transfer and external utility. All intervention routes first require controlled brain predictivity and quality-aware headroom. Routes using synthetic brain responses require one additional prerequisite: the exact synthetic target must predict independently recorded fMRI responses beyond nuisance features and the declared frozen-generator control; routes using recorded fMRI responses as training targets bypass this check.

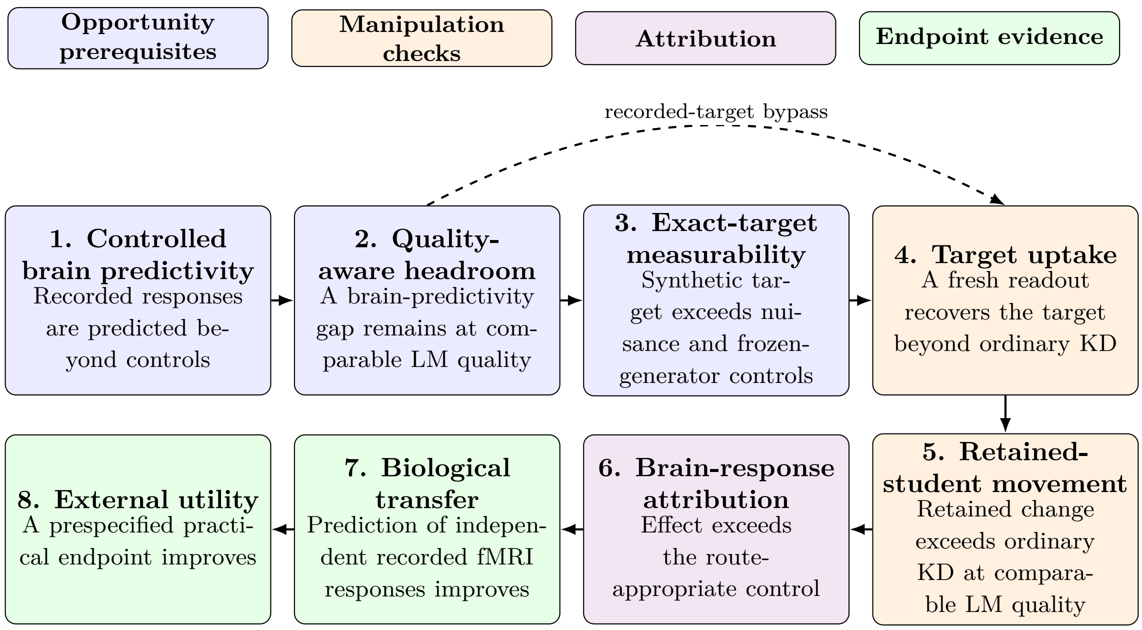

Figure 1: Evidence chain for brain-response-specific training claims.

No single comparator supports every stage in Figure 1. At stage 1, an architecture-matched untrained LM tests whether controlled brain predictivity depends on learned parameters. At stages 4 and 5, an ordinary-KD arm isolates the incremental effect of adding a brain-response objective to the same student-training regime, while comparable LM quality prevents generic quality differences from confounding the comparison. At stage 6, the route-appropriate attribution control depends on the training target. The recorded-response route compares correctly paired with permuted responses, testing whether the stimulus–response pairing matters while preserving the response values and their marginal distribution. The synthetic-response route compares the synthetic brain-response target with a text-derived auxiliary target matched in geometry, scale, baseline predictability, learnability, and optimization pressure. For either route, brain-response-specific attribution requires the brain-response-target arm to outperform its control on the outcome being claimed. At stage 7, a freshly fitted readout on independently recorded fMRI responses tests biological transfer; successful recovery of the optimized training target cannot substitute for this endpoint. Stage 8 requires improvement over the appropriate baseline on a prespecified practical endpoint.

Stages 1–7 are conjunctive for a claim of brain-response-specific improvement in predicting recorded fMRI responses; stage 8 is additionally required only for a practical-utility claim. Each stage licenses only its stated claim. If an earlier stage fails or remains unresolved, later target uptake, retained-student movement, or endpoint results may still be reported descriptively, but they cannot establish attribution or transfer that depends on the earlier stage. Conversely, failure at a later stage does not erase evidence supporting an earlier, narrower claim. An unresolved comparison must remain unresolved rather than be converted into a null result, and failure under one readout, cohort, or target must not be generalized beyond that scope. This conjunctive interpretation prevents evidence at a later stage from substituting for an unsupported earlier claim and identifies the stage that requires redesign. Section 3 maps each stage to its data, training arms, quantity or effect being estimated, comparator, endpoint, and inference unit.

3 Experimental Design and Evaluation

Section 2.4 defines the evidence required to convert controlled brain predictivity into a brain-specific training claim. This section instantiates that evidence standard using distinct data, interventions, assays, comparators, endpoints, and inference units. An estimand is the specific quantity or effect that an analysis is designed to estimate. Table 2 maps each conceptual stage to its thesis-facing scientific role and corresponding design tuple. A stage may appear in more than one row when different datasets, response constructions, or population units define different estimands. This section specifies the study design; Section 4 reports the verdict for each role.

Table 2: Scientific roles within the evidence chain.

Stage	Scientific role	Primary comparison and endpoint	Inference unit
1	Controlled regional predictivity assay	Trained representations versus nuisance and architecture-matched untrained LM; held-out ordinary and unique R^2	Fixed sentence set; five contiguous folds
1	Controlled naturalistic voxelwise predictivity assay	Trained versus untrained representations beyond nuisance; held-out-story voxelwise unique R^2	Story blocks within one deeply sampled participant
2	Quality-aware distillation-headroom analysis	Declared-layer brain predictivity before and after ordinary KD, interpreted with held-out LM quality	Model condition; training seed for stochastic arms
3	Exact synthetic-target measurability assay	Synthetic target versus nuisance and frozen row twin; held-out recorded-response unique R^2	Participant on a fixed sentence set
4	Synthetic-target recovery assay	Synthetic-response versus ordinary-distillation arms; saved block-permuted variants are sensitivities	Training seed
5	Retained-student movement audit	Synthetic-response versus ordinary-distillation arms; retained parameter or representation change at comparable LM quality	Training seed; fixed target sample for geometry
6	Recorded-response intervention, participant-averaged estimand	Correctly paired versus permuted response targets; fresh averaged-response unique R^2	Shared stimulus fold; seeds are technical repetitions
6	Recorded-response intervention, participant-specific estimand	Correctly paired versus permuted response targets; fresh participant-response unique R^2	Participant after fold and seed aggregation
6	Comparator-adequacy audit and attribution assessment	Synthetic-response versus text-derived auxiliary-control arms; target parity, learnability, retained movement, and downstream arm contrasts	Seed for trained models; fixed target sample for matching checks
4	Saved-student target-retention analysis	Synthetic-response versus ordinary-distillation students; held-out synthetic-target recovery	Training seed
7	Composed-path diagnostic	Aligned versus frozen row-twin targets and synthetic-response versus ordinary-distillation students; held-out recorded-response unique R^2 through the composed path	Participant
7	Saved-student biological-transfer assay	Synthetic-response versus ordinary-distillation and text-derived auxiliary-control arms; fresh participant-response unique R^2	Participant
8	Bounded practical-utility evaluation	Brain-response-target arm versus route-appropriate quality-matched control; held-out language or task endpoint	Technical seed; fixed corpora and procedure

3.1 Study Design, Datasets, and Data Partitioning

Two datasets provide the recorded fMRI responses used in this study. The Tuckute condition-B dataset provides the sentence-level substrate [23]. It contains 1,000 isolated English sentences, responses from 10 participants, and five fixed left-hemisphere language regions of interest (ROIs). The dataset supports the controlled regional predictivity assay and the quality-aware distillation-headroom analysis. The same sentence-level substrate later supports the recorded-response route and the Tuckute endpoints for exact synthetic-target measurability and saved-student biological transfer. The LeBel corpus provides the naturalistic substrate for the controlled naturalistic voxelwise predictivity assay [24]. That assay uses participant UTS03 and 20 stories divided into five held-out-story folds. A separate repeated story determines which voxels meet the reliability criterion. Holding out complete stories prevents temporally dependent samples from the same story from crossing the evaluation boundary. The resulting estimand concerns held-out-story prediction within UTS03 and does not support population inference across people.

The Tuckute responses can be constructed at two response-aggregation levels. Participant averaging combines the five selected participants into one sentence-by-ROI matrix. This matrix describes the shared stimulus-locked response, but it cannot support participant-level inference. Participant-specific construction begins from the 10 recorded participants. One participant is excluded because the required five-ROI response matrix of the participant is incomplete, leaving nine valid participants. Each remaining participant retains a separate response matrix. Contrasts can therefore be aggregated within participant before inference across the nine tested participants. Spatial resolution is a separate distinction. An ROI coordinate summarizes responses within a predefined language region, whereas a voxel coordinate preserves a finer spatial measurement. Section 3.5 pairs these constructions with their estimands and inference units.

Each training and evaluation role draws on a declared text and response source. The recorded-response intervention trains on Tuckute sentences and their recorded responses. The synthetic-response intervention instead trains on 95,999 WikiText-103 sentences [25]. A separate 1,999-sentence WikiText split supports language-quality evaluation and synthetic-target recovery. For the exact synthetic-target measurability assay, the fixed target is generated solely from the Tuckute sentence text. Recorded fMRI responses enter only its held-out evaluation. Saved students then remain frozen while fresh readouts evaluate biological transfer to those responses. Pereira sentences and LeBel story transcripts provide out-of-domain language-quality endpoints for the bounded practical-utility evaluation [24, 26]. Recorded LeBel fMRI responses are confined to the controlled naturalistic voxelwise predictivity assay.

The data partitions preserve these separate roles. The principal Tuckute roles use five contiguous outer partitions. Each partition holds out 200 sentences and leaves the other 800 as its training complement. For the controlled regional predictivity assay, quality-aware distillation-headroom analysis, exact synthetic-target measurability assay, and saved-student biological-transfer assay, this boundary separates fresh-readout fitting from evaluation. The recorded-response intervention uses the outer boundary differently. It optimizes the student on the 800-sentence training portion, which excludes the held-out 200 sentences. Within that held-out block, five inner folds separate fresh-readout fitting from readout evaluation. The synthetic-response route uses its two WikiText partitions for student training and target-recovery evaluation. Within that route, recorded Tuckute responses enter only the later exact synthetic-target measurability and saved-student biological-transfer assays. Fold-dependent transformations are fitted on the corresponding training portion. The exact synthetic-target measurability assay has one declared exception: it reuses target-standardization moments fixed externally during target construction. Its subsequent target principal component analysis (PCA) and all readout transformations remain fold-local. Appendices B and D report the exact inclusion, preprocessing, partition, and nesting definitions.

Figure 2 condenses these assignments into three routes.

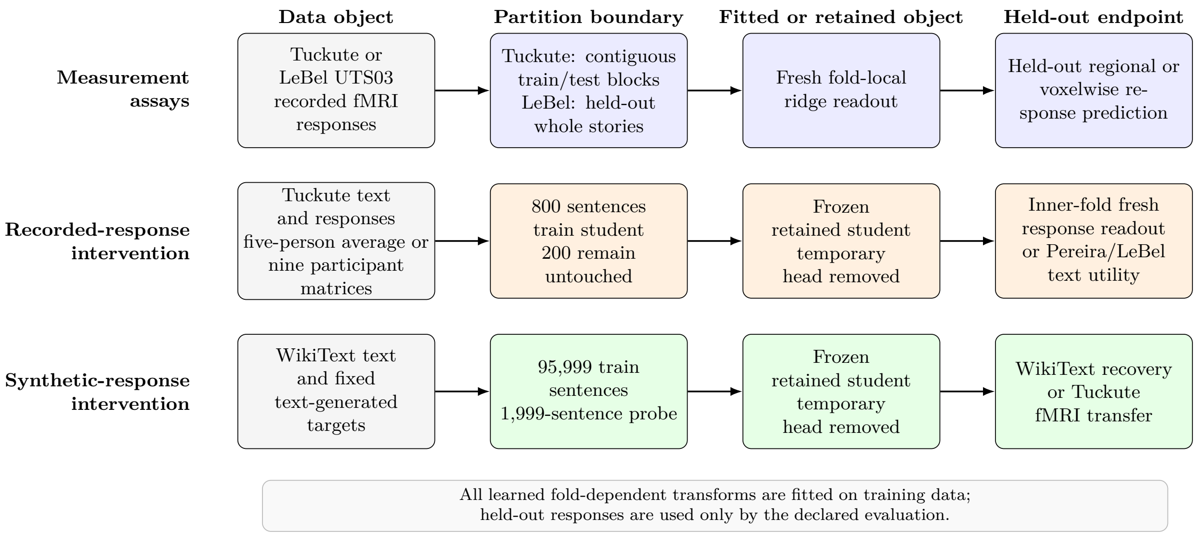

Figure 2: Data and partition routes used by the Section 3 designs.

The measurement lane fits readouts without changing an LM; the two intervention lanes first produce a retained student and then evaluate it on a held-out target. The arrows denote data use, training, or evaluation rather than causal influence.

3.2 Measurement Prerequisites

Both controlled predictivity assays begin with the same question: does a contextual LM representation improve held-out prediction of recorded fMRI responses beyond specified nuisance features? Each assay first fits a nuisance-only readout containing its declared low-level and static features. The full readout adds a contextual representation block. The contextual and static blocks are reduced to the same PCA rank, which matches their individual widths. Let Y_j denote response coordinate j, N the nuisance block, X the contextual block, and $\mathcal{T}$ one held-out fold. For any readout design M, ordinary held-out R^2 is

$$R_j^2(M) = 1 - \frac{\sum_{i\in\mathcal{T}}(y_{ij} - \widehat{y}_{ij}^{M})^2}{\sum_{i\in\mathcal{T}}(y_{ij} - \bar{y}_{\mathcal{T}j})^2}. \quad (1)$$

Equation 1 expresses held-out performance as the fraction of response variation explained beyond a prediction that always uses the held-out mean. A value of one denotes perfect prediction, zero matches that mean baseline, and a negative value means that the fitted readout predicts worse than the baseline. The nuisance-only and full scores are $R_j^2(N)$ and $R_j^2([N, X])$. Their cross-validated semipartial, or unique, score is

$$\Delta R_j^2(X) = R_j^2([N, X]) - R_j^2(N). \quad (2)$$

Equation 2 measures the change in held-out predictive value when the contextual block is added to the declared nuisance design. A positive value therefore means that the contextual LM representation improves held-out response prediction beyond those nuisance features.

Figure 3 separates this nuisance comparison from the two additional controls. One control uses an untrained LM with the same architecture and rank. It tests whether learned parameters matter. The frozen row twin instead tests whether correct sentence–target pairing matters. Repeating the contextual comparison across the teacher and students supplies the scores used by the quality-aware headroom analysis.

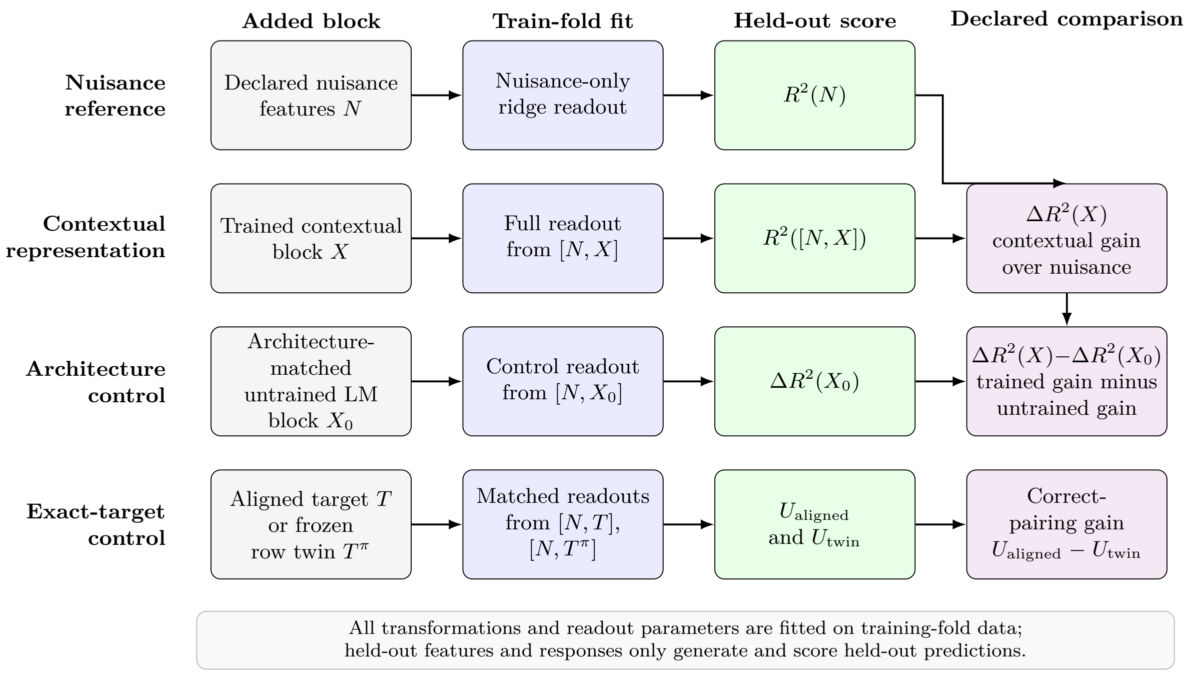

Figure 3: Fold-local readout comparisons used by the measurement assays.

The test-fold mean $\bar{y}_{\mathcal{T}j}$ supplies the reference prediction in Equation 1. Both $R^2_j(M)$ and $\Delta R^2_j(X)$ can be negative. For Tuckute, these readouts use contiguous sentence folds, and released reliability estimates provide descriptive context. For LeBel, they hold out complete stories, and a separate repeated story determines voxel eligibility.

The controlled assays define the measurement prerequisite, but they do not show how much brain predictivity ordinary KD leaves available to recover. The quality-aware distillation-headroom analysis addresses that next question by applying the same Tuckute assay at the declared layers. Its reference models are the pretrained `gpt2-medium` teacher, an architecture-matched untrained `gpt2`, and an off-the-shelf pretrained `gpt2`. The analysis also evaluates two `gpt2` students after ordinary KD. One student begins from random initialization. The other begins from pretrained weights. Let $\overline{\Delta R^2}(m, s)$ denote the declared-ROI and fold average of Equation 2 for model m and technical seed s. For teacher m_T, the seed-specific headroom is

$$\mathcal{H}(m, s) = \overline{\Delta R^2}(m_T) - \overline{\Delta R^2}(m, s). \tag{3}$$

Equation 3 measures the teacher predictivity that the tested student has not recovered. Positive headroom means that the teacher retains a predictivity advantage, while zero headroom means that their averaged scores are equal under this assay. The seed index is omitted for deterministic reference models. Each brain-predictivity score is interpreted with held-out LM quality because quality may explain part of the difference. Perplexity provides this comparison within the shared-tokenizer study. A separate cross-family analysis uses tokenizer-comparable bits per byte. That cross-family analysis relates $\overline{\Delta R^2}(m)$ directly to model quality; it does not define teacher-relative headroom across unrelated model families. The analysis measures headroom under the tested conditions. It does not identify model size, architecture, initialization, training history, distillation, or language quality as the cause of that headroom.

The exact synthetic-target measurability assay asks whether the fixed synthetic brain-response target generated by TRIBE from text predicts recorded Tuckute responses on the same 1,000 sentences. For each participant, the assay fits three otherwise matched response readouts. The first contains nuisance features only. The second adds the aligned synthetic target. The third adds its frozen row-twin control. This control applies one fixed permutation to complete target rows within each outer block and has no fixed points. It preserves the exact target-row multiset within that block but breaks sentence–target pairing. Let T denote the aligned target block and T^π its frozen row twin. Both use the same train-fold target-PCA basis and the same nuisance and readout procedure. Write $R^2_{u,f,j}(M)$ for Equation 1 applied to participant u, fold f, and response coordinate j. For the J declared response coordinates, define

$$\begin{aligned} U_{\text{aligned}}(u,f) &= \frac{1}{J}\sum_{j=1}^{J}\left[R^2_{u,f,j}([N,T]) - R^2_{u,f,j}(N)\right], \\ U_{\text{twin}}(u,f) &= \frac{1}{J}\sum_{j=1}^{J}\left[R^2_{u,f,j}([N,T^\pi]) - R^2_{u,f,j}(N)\right]. \end{aligned} \tag{4}$$

These two quantities apply the same nuisance subtraction to the correctly paired target and its row-twin control. Their difference can therefore test whether correct sentence–target pairing adds predictive value beyond the information retained by the target rows after permutation.

Let $A(u)$ denote participant-level measurability beyond nuisance. Let $Q(u)$ denote the additional predictive value of correct sentence–target pairing. The two participant-level estimands are

$$\begin{aligned} A(u) &= \frac{1}{K}\sum_{f=1}^{K} U_{\text{aligned}}(u,f), \\ Q(u) &= \frac{1}{K}\sum_{f=1}^{K}\left[U_{\text{aligned}}(u,f) - U_{\text{twin}}(u,f)\right], \qquad K=5. \end{aligned} \tag{5}$$

Equation 5 first averages the aligned target's unique predictivity across folds and then averages the aligned-minus-twin pairing contrast across those same folds. Computing both quantities separately for each participant makes the participant, rather than a fold or response coordinate, the biological inference unit. The nine pairs $\{A(u), Q(u)\}$ enter participant-level inference. The assay therefore tests participant-level predictive measurability on this fixed stimulus set. It does not establish causal mediation, brain-response-specific attribution, or stimulus-population generality.

All measurement prerequisites use the same general held-out readout discipline. A fresh ridge readout is fitted to frozen features for every evaluated model or target. Ordinary R^2 compares prediction error with variation in the held-out responses. Negative values are retained when predictions are worse than the held-out-mean baseline. Equations 1 and 2 apply across the prerequisite assays. All learned preprocessing, response centering, regularization selection, and readout fitting follow the training-partition rule in Section 3.1. Appendix B specifies the exact inputs, preprocessing, and declared standardization exception. Appendix D specifies the formulas, aggregation, and inference procedures.

3.3 Recorded-Response Intervention

Within each outer fold and technical seed, the recorded-response and permuted-response arms start from the same student state and use the same sentences. They also share the ordinary-KD objective, optimizer schedule, loss weight, and training budget. Only the pairing between recorded-response rows and their sentences differs. Both arms distil a Qwen2.5-1.5B teacher into a Qwen2.5-0.5B student using low-rank adaptation (LoRA). The teacher remains frozen. Using the partitions defined in Section 3.1, each run trains on 800 Tuckute sentences and leaves the same contiguous block of 200 sentences untouched. Five outer folds and three technical seeds repeat this paired design.

Ordinary KD provides the shared language-preservation objective. Let $\mathcal{L}_{\text{KD}}$ denote the implemented temperature-scaled, teacher-to-student Kullback–Leibler (KL) divergence over all nonpadding tokens. Let $\mathcal{L}^{(a)}_{\text{R}}$ denote the auxiliary response loss for arm a. The complete training objective is

$$\mathcal{L}_{\text{train}}^{(a)} = \mathcal{L}_{\text{KD}} + \lambda_{\text{R}}\mathcal{L}_{\text{R}}^{(a)}, \qquad a \in \{\text{recorded}, \text{permuted}\}. \tag{6}$$

Equation 6 says that both paired arms are anchored by the same teacher-defined language objective while receiving an additional response-based training signal. The arm identity changes the response pairing, not the language objective or the strength assigned to the response loss. For a minibatch $\mathcal{B}$, let $\bar{h}_i^{(12)}$ be the masked-mean layer-12 student state, W a temporary linear map, $d = 5$ the number of response coordinates, and $y_i^{(a)}$ the target for sentence i. The auxiliary loss is

$$\mathcal{L}_{\text{R}}^{(a)} = \frac{1}{|\mathcal{B}|d}\sum_{i\in\mathcal{B}} \left\| W\bar{h}_i^{(12)} - y_i^{(a)} \right\|_2^2. \tag{7}$$

Equation 7 measures how far the temporary head's prediction from the student's layer-12 representation is from the assigned response target. Minimizing it can change the retained student only when its gradient reaches trainable parameters that produce that representation. The primary intervention uses $\lambda_{\text{R}} = 10$. The ordinary-KD reference sets $\lambda_{\text{R}} = 0$, but it remains secondary because its language quality need not match that of the response-target arms. Appendix C specifies the exact temperature, masking reduction, optimizer, schedule, and parameter states.

The recorded-response arm uses the response vector paired with each sentence. The control applies the same loss to block-permuted response rows. This permutation preserves the response values, their marginal distribution, and their within-block structure while disrupting the sentence–response correspondence. Because a random block permutation can leave some blocks in their original positions, it is a matched permutation control rather than an exact derangement. The participant-specific assay repeats this control with five permutation draws and averages their evaluated scores when forming the paired contrast. Appendix B gives the exact permutation construction.

Participant-averaged and participant-specific responses instantiate the same paired intervention but answer different questions. In the participant-averaged regime, each fold–seed run uses the fixed cellwise participant-averaged response construction defined in Section 3.1. Its contrast describes the effect of correct pairing for that shared stimulus-locked target. The participant-specific regime instead trains separate paired arms for each of the nine participants with complete five-ROI response matrices. Its contrast supports inference about variation across the tested participants. Averaging responses before training is not equivalent to averaging participant effects after training. The participant-averaged trend therefore cannot be interpreted as the cohort-mean intervention effect.

The auxiliary branch attaches to the masked-mean layer-12 student state through the temporary head W. In the primary LoRA regime, the pretrained base model and tied input/output embedding remain frozen, while W and the adapters are optimized. The auxiliary gradient can reach W and trainable adapters that produce the layer-12 state. It cannot reach later adapters through this branch, although those adapters can still receive gradients from $\mathcal{L}_{\text{KD}}$. After training, the temporary head and teacher are discarded, the adapters are merged into the retained student, and that student is frozen for evaluation. Figure 4 separates the shared training core from frozen-student evaluation. Panel a shows where the ordinary-KD and route-specific auxiliary objectives attach. Panel b shows that every post-training endpoint uses the retained student rather than the temporary target head.

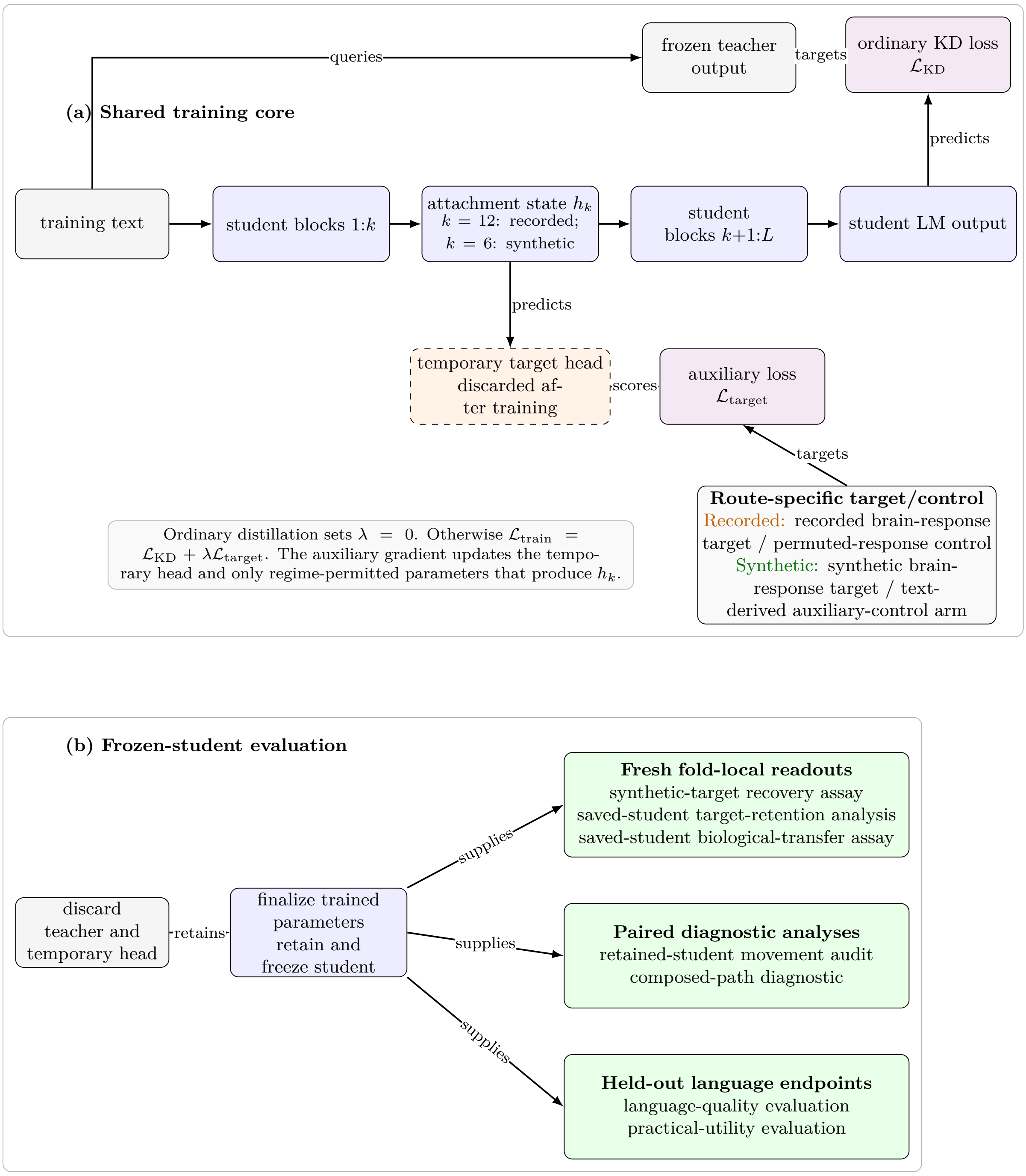

Figure 4: Shared intervention training and frozen-student evaluation paths.

After each run, evaluation uses only the untouched 200-sentence block. A fresh ridge readout is fitted and tested within that block using five contiguous inner subfolds. Its endpoint is the ROI-averaged unique R^2 defined by Equation 2. Let $U_{ufs}^{(a)}$ denote this score for participant u, outer fold f, technical seed s, and arm a. With $P = 5$ permutation draws, the paired cell contrast and participant aggregate are

$$\begin{aligned} d_{ufs} &= U_{ufs}^{(\text{recorded})} - \frac{1}{P}\sum_{p=1}^{P} U_{ufsp}^{(\text{permuted})}, \\ e_u &= \operatorname{median}_{f,s} d_{ufs}. \end{aligned} \tag{8}$$

Equation 8 first compares the recorded-response student with the average of its five matched permutation controls within each fold–seed cell. It then summarizes those technical repetitions within each participant, so one unusually large fold or seed cannot by itself define that participant's effect. The participant-level point estimate is

$$\widehat{\delta}_{\text{participant}} = \frac{1}{9}\sum_{u=1}^{9} e_u. \tag{9}$$

Equation 9 averages the nine participant effects to estimate the intervention contrast in the tested cohort. The nine values e_u, rather than the underlying seeds or folds, provide the biological inference values. A separate aggregation across the five shared outer folds checks whether the contrast is concentrated in particular stimuli. The participant-averaged analysis instead aggregates seeds within each shared fold and remains descriptive of its fixed response target. A held-out WikiText probe checks language-quality balance between arms. Appendix D specifies the intervals, tests, shared-fold checks, and sensitivity analyses.

The frozen auxiliary-head, increased-LoRA-capacity, contrastive-objective, naturalistic voxelwise, and full-fine-tuning studies retain the paired recorded-versus-permuted comparison while changing the target, target loss, language anchor, trainable subset, or evaluation scope. Appendix C defines these variants, and their study-specific outcomes are reported in the Results. None replaces the primary nine-participant contrast.

3.4 Synthetic-Response Intervention and Downstream Assays

The synthetic-response intervention asks whether fixed, text-generated brain-response targets change a retained student beyond ordinary KD. Across six technical seeds, a frozen GPT-2 Medium teacher supervises a GPT-2 student in the synthetic-response, ordinary-distillation, and text-derived auxiliary-control arms. These arms share the initialized student, training text and order, tokenizer, ordinary-KD objective, optimizer schedule, and training budget. The synthetic-response arm adds a fixed text-only TRIBE target, whereas the text-derived auxiliary-control arm mean-pools frozen teacher states and maps them through a fixed Gaussian projection. Both auxiliary targets are standardized to 20,484 coordinates and predicted from the student's layer-6 representation with a mean-squared-error loss. Training updates the student and a temporary prediction head; the head is then discarded and the retained student is frozen. Saved block-permuted variants provide sensitivity checks rather than peer claim-identifying arms. In Figure 4, panel a locates the synthetic target and text-derived auxiliary control within the shared training core. Panel b separates target recovery, retained movement, biological transfer, composed predictive overlap, and language evaluation after the temporary head has been removed. Appendices B and C specify the target construction and exact training procedure.

Two manipulation checks ask whether the intervention affected the retained student before any downstream claim is considered. First, the synthetic-target recovery assay fits a fresh readout to frozen student representations on held-out WikiText examples and averages target-coordinate R^2 with equal weight across coordinates. Recovery above the ordinary-distillation reference shows that the retained representation makes more of the training target linearly accessible after the temporary head has been removed. Second, the movement audit compares each trained student with its ordinary-distillation counterpart through parameter displacement, centered relative-Frobenius distance, and layer-6 centered kernel alignment (CKA). These measurements establish retained change, but they do not identify which property of the auxiliary target caused it. The six seeds quantify technical variation rather than biological uncertainty. This WikiText recovery assay is distinct from the Tuckute target-retention analysis described below, which uses nuisance-controlled, variance-weighted response-space scoring.

Brain-response-specific attribution additionally requires an adequate text-derived comparator. The comparator audit groups its checks by the source of a possible alternative explanation. Training parity covers the shared optimization conditions and the initial auxiliary-loss and gradient scales. Target parity covers scale, covariance, effective rank, row order, and nuisance predictability. Baseline learnability compares ordinary-distillation recovery and remaining headroom, while the movement checks compare parameter and representation changes induced by training. Passing every required axis is necessary but not sufficient for attributing an effect to brain-response content. Failure on any required axis makes the contrast non-identifying; if no required axis fails but a required measurement is missing, attribution remains unresolved. The available block-permuted targets are also non-identifying because their construction can retain aligned rows and can duplicate or omit a row. A frozen Sattolo row twin therefore supplies an exact target-only sensitivity. Because no student was trained against that twin, it cannot replace the text-derived trained-arm comparator.

The saved-student biological-transfer assay then asks whether the synthetic-response intervention improves prediction of independently recorded responses. Let $U^{(a)}_{ufs}$ be the nuisance-controlled, ROI-averaged unique R^2 for participant u, contiguous fold f, technical seed s, and training arm a. The direct biological-transfer endpoint uses the ordinary-distillation arm as its reference. With $F = 5$ folds and $S = 6$ seeds, define

$$
\begin{aligned}
d^{\text{direct}}_{ufs} &= U^{(\text{synthetic})}_{ufs} - U^{(\text{KD})}_{ufs}, \\
e^{\text{direct}}_{u} &= \frac{1}{FS}\sum_{f=1}^{F}\sum_{s=1}^{S} d^{\text{direct}}_{ufs}, \\
\widehat{\delta}_{\text{direct}} &= \frac{1}{9}\sum_{u=1}^{9} e^{\text{direct}}_{u}.
\end{aligned}
\tag{10}
$$

Equation 10 first compares paired synthetic-response and ordinary-distillation students within each fold–seed cell, then averages technical repetitions within each participant. The final mean therefore describes biological transfer across the nine tested participants, with participants rather than folds or seeds as the biological inference units. All saved students remain frozen while fresh ridge readouts are fitted within the five contiguous folds over the 1,000 Tuckute sentences; layer 7 is primary and layer 6 is a prespecified sensitivity analysis. The direct contrast answers whether the synthetic intervention improves recorded-response predictivity relative to ordinary distillation, without attributing any difference to brain-response content.

The preregistered attribution endpoint uses the same cell scores but changes the comparator. The arm label textctrl denotes the text-derived auxiliary-control arm. Its cell contrast is

$$
\begin{aligned}
d^{\text{control}}_{ufs} &= \left(U^{(\text{synthetic})}_{ufs} - U^{(\text{KD})}_{ufs}\right) - \left(U^{(\text{textctrl})}_{ufs} - U^{(\text{KD})}_{ufs}\right) \\
&= U^{(\text{synthetic})}_{ufs} - U^{(\text{textctrl})}_{ufs}.
\end{aligned}
\tag{11}
$$

Equation 11 expresses the difference between two gains over ordinary distillation and shows why it reduces to synthetic response minus text-derived control. The cancellation is valid because the corresponding ordinary-distillation baselines were verified to be byte-identical within each seed. This contrast was preregistered as the primary attribution comparison, but its interpretation remains conditional on comparator adequacy.

The same saved students support a second, mechanism-localizing analysis. Let $A_{H\to T|N}$ denote the representation-specific coefficient contribution in the train-fold model that predicts the synthetic target from nuisance features N and a frozen student representation H. Let $A_{T\to Y|N}$ denote the target-specific coefficient contribution in the train-fold model that predicts recorded responses Y from N and the target T, and let $\widehat{Y}_N$ be the nuisance-only response prediction. The composed prediction is

$$
\widehat{Y}_{\text{comp}} = \widehat{Y}_N + A_{T\to Y|N}\left(A_{H\to T|N}(H_{\text{test}})\right), \tag{12}
$$

Equation 12 asks how much held-out recorded-response predictivity can pass through the specific synthetic-target subspace recovered from a frozen student. Both conditional coefficient contributions and all fold-dependent preprocessing transforms are estimated from training-fold data only. The externally fixed moments from the synthetic-target construction first standardize the generated target. Before response mapping, this contribution is converted with the train-fold target scale retained from the target-to-response model. Comparing aligned and Sattolo-row-twin targets tests whether sentence-level target pairing matters, while comparing synthetic-response and ordinary-distillation students tests whether training changed access to that path. This composed score measures predictive overlap; it does not establish causal mediation. The direct saved-student endpoint remains the primary biological-transfer endpoint.

Language-quality checks qualify the synthetic-response contrasts but do not themselves establish practical benefit. Perplexity and bits per byte test whether the compared students remain within the declared quality margin. Passing the 0.005 bits-per-byte margin only excludes discrepancies larger than that margin on this metric; it does not prove the absence of language-quality confounding. A separate practical-utility evaluation belongs to the recorded-response route. It compares approximately quality-matched recorded-response and permuted-response students through out-of-domain to in-domain perplexity ratios on Pereira sentences and LeBel story text. Ordinary-distillation and text-derived-control arms remain descriptive references when their language quality is unmatched. This utility contrast is interpretable only if the recorded-minus-permuted alignment mediator is reliable relative to its measured minimum detectable effect. Its seeds quantify technical variation, not independent biological evidence. Section 3.5 states the formal aggregation and decision rules.

3.5 Cross-Cutting Estimands and Decision Rules

Table 2 binds each scientific role to its estimand, comparator, endpoint, and inference unit. The focal and comparator conditions are first evaluated on the same endpoint within the smallest matched analysis cell. Measurement roles compare representation-based predictions with nuisance or untrained controls. Intervention roles compare paired targets or training arms. Transfer and utility roles require an independently evaluated biological or language endpoint. A numerical difference changes meaning when its comparator, endpoint, or inference unit changes, even if the same scoring function is used.

Figure 5 shows the common order from a matched analysis cell to the declared inference unit. Its three tracks separate biological inference across participants, fixed-stimulus inference across shared folds or stories, and technical inference across seeds. The eligibility check qualifies interpretation but neither changes the inference unit nor demonstrates an endpoint.

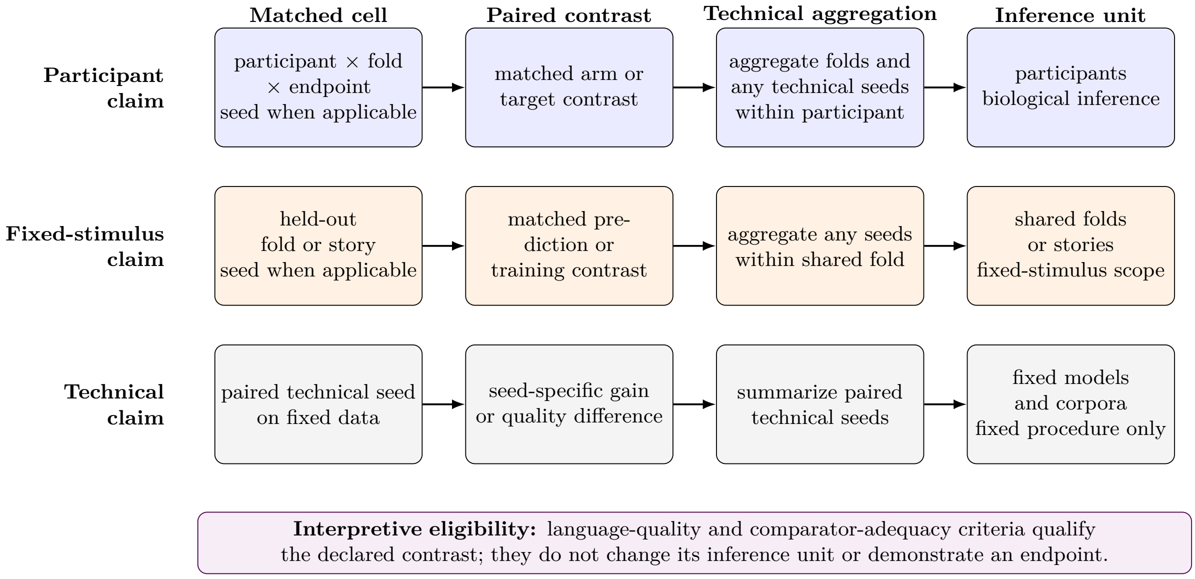

Figure 5: Aggregation paths for participant, fixed-stimulus, and technical claims.

A training contrast becomes eligible for interpretation only under its declared LM-quality rule. The quality metric follows the declared assay. The scaled synthetic-target family and recorded-response probes use perplexity under a shared tokenizer and scoring protocol. Cross-tokenizer or cross-family comparisons require bits per byte, and the saved-student transfer assay uses its preregistered bits-per-byte criterion even though its models share a tokenizer. Let $\mathrm{PPL}_{a,s}$ be the held-out perplexity of target arm a in seed s. Let $L_{a,s}$ and $B_{a,s}$ be its total negative log-likelihood in bits and the number of decoded UTF-8 bytes in the same scored span. Let $\epsilon_{\mathrm{ppl}} = 0.05$ and $\epsilon_{\mathrm{bpb}} = 0.005$ be the declared relative-perplexity and bits-per-byte tolerances, respectively. The two principal quality criteria are

$$\begin{aligned} q_{\mathrm{ppl}}(a,s) &= \frac{|\mathrm{PPL}_{a,s} - \mathrm{PPL}_{\mathrm{KD},s}|}{\mathrm{PPL}_{\mathrm{KD},s}} \leq \epsilon_{\mathrm{ppl}}, \\ q_{\mathrm{bpb}}(a,b,s) &= \left| \frac{L_{a,s}}{B_{a,s}} - \frac{L_{b,s}}{B_{b,s}} \right| \leq \epsilon_{\mathrm{bpb}}. \end{aligned} \tag{13}$$

The first rule requires each scaled synthetic-target arm to remain within 5% of its seed-matched ordinary-KD reference on the fixed WikiText quality probe. The second requires the saved-student comparisons to remain within 0.005 bits per byte in every seed. Passing a rule makes the contrast eligible under that metric and tolerance; it does not prove identical language behavior or remove possible differences inside the accepted margin. The paired recorded-response and permuted-response arms are evaluated on the same fixed WikiText probe and must satisfy their declared approximate quality match. Recorded-response comparisons with ordinary KD additionally require checkpoint or curve matching on perplexity. Without the required match, the corresponding biological contrast remains descriptive.

Aggregation follows the claim rather than the largest available number of observations. The participant-averaged recorded-response estimand averages seeds within each outer fold and treats the five shared folds as its fixed-stimulus inference values. The participant-specific recorded-response estimand instead aggregates seed–fold cells within each person before inference across participants. It uses the declared within-participant median. Saved-student biological transfer averages six seeds and five folds within each participant before inference across the nine participants. Within the exact-substrate diagnostic, exact synthetic-target measurability averages folds within participant and has no student-seed axis. The composed-path diagnostic averages folds and seeds within participant, whereas Tuckute target retention pools SSE/SST and summarizes six paired technical-seed contrasts. Synthetic-target recovery uses six matched technical-seed gains and does not provide biological replication. The WikiText recovery assay averages coordinate-level R^2 equally, whereas the Tuckute target-retention diagnostic pools SSE/SST across coordinates. These two aggregations therefore answer different target-recovery questions.

The inference unit also fixes the scope of uncertainty. The controlled sentence assays report held-out-fold dispersion on a fixed sentence set. The naturalistic assay retains a within-participant trained-minus-untrained contrast and the signs of its five held-out story-block differences. Its earlier voxel-bootstrap interval is excluded because spatially dependent voxels from one participant are not independent population observations. Quality-aware headroom uses a fold–seed bootstrap conditional on convergence, while cross-family quality relations use family-cluster bootstrap estimates with Fisher intervals. The participant-averaged recorded-response intervention uses a five-fold t interval after technical seeds have been aggregated within fold. Synthetic-target uptake uses paired gains over six technical seeds, descriptive seed intervals, and the declared exact magnitude-preserving sign-flip test. Coordinates, voxels, folds, and seeds remain assay or technical measurements rather than substitutes for participants.

When the intended claim concerns people, uncertainty must be computed at the participant level after all technical aggregation. The participant-specific recorded-response analysis reports a participant t interval and an exact one-sided sign test. A separate shared-fold t interval and cluster bootstrap assess whether its direction is stable across stimulus blocks; both participant and shared-fold readings must support a population claim. The saved-student biological-transfer assay reports a participant t interval, a one-sided upper bound, an exact sign test, and a Wilcoxon signed-rank sensitivity, while its seed and block summaries remain descriptive. Exact synthetic-target measurability and the composed-path diagnostic use participant intervals and exact signs. Tuckute target retention instead uses a descriptive paired-seed interval and an exact 2^6 sign-flip test. Separate Bonferroni family-95% intervals and simultaneous bounds govern target measurability, target retention, and the composed student-to-target-to-response path. The practical-utility evaluation instead uses eight technical seeds for paired arm contrasts and therefore remains bounded to its fixed models, corpora, and procedure. Appendix D gives the exact estimators, nesting, tests, and sensitivity variants.

Detectable-effect and continuation bounds answer narrower questions than statistical intervals. An minimum detectable effect (MDE) describes the effect size that an assay could reliably resolve under its stated variance model. It does not imply that smaller effects are zero or biologically irrelevant. The previous naturalistic follow-up calculation is excluded because it used unmatched absolute fold variability rather than the intended paired contrast. The participant analyses calculate detectable-effect bounds from between-participant variation, whereas the practical-utility analysis uses variation across technical seeds. The inherited $+0.002$ unique-R^2 reference is a fixed continuation rule for the declared participant assay. It is not a universal biological-importance threshold and cannot be transferred to another response scale, nuisance model, layer, or population.

Evidence status follows the prerequisite chain. A passed manipulation check establishes target uptake or retained-student movement, but it does not establish brain-response-specific attribution, biological transfer, or practical utility. Failure of a required manipulation or quality criterion stops the downstream contrast from carrying its intended claim. Comparator adequacy is a necessary conjunction: failure on any required axis makes the attribution contrast non-identifying. An unmeasured required axis leaves adequacy unresolved only when no measured required axis has already failed. An unmeasured stage supplies no evidence and is not a null result. The label *not demonstrated* is reserved for an identifying, adequately measured contrast that does not pass its decision rule. The Results apply these statuses in scientific-role order rather than experiment number or execution date.

4 Results

The evidence chain separates a measured signal from a successful intervention. Table 3 uses *supported* when the declared stage passes, *not demonstrated* when the tested contrast does not support its target claim, *unresolved* when the evidence cannot decide, and *non-identifying comparator* when the available control cannot support attribution to brain-response content. These labels are scoped to the estimand, assay, and inference unit stated in Sections 3.1–3.5.

Table 3: Verdict-first status of the evidence chain. The table reports the strongest evidence and its licensed interpretation, not the experiment chronology.

Stage	Status	Strongest evidence	Interpretation
Controlled brain predictivity	**Supported**	Trained representations exceed architecture-matched untrained controls on both recorded-fMRI substrates; naturalistic gaps are +0.0207 and +0.0277 .	Controlled predictivity exists under the tested linear assays; no intervention or population claim follows.
Quality-aware headroom	**Supported**; source **unresolved**	Cold-start KD leaves a teacher gap of +0.0181 and floor-anchored retention of 0.37 [0.14, 0.58] .	There is observed room to intervene, but language quality competes with an objective-specific explanation .
Recorded-target pairing	**Not demonstrated**	Mean of nine participant medians +0.00010, with participant t interval [−0.00037, +0.00058] .	The averaged-target trend does not become a participant-general advantage.
Exact-target measurability	**Not demonstrated**; twin **unresolved**	Aligned-target increment −0.004494, simultaneous 95% interval [−0.010695, +0.001707]; aligned-minus-twin contrast remains uncertain .	The predicted-target branch misses its first frozen biological prerequisite.
Target uptake and retained-student movement	**Supported** on WikiText continuity; fresh Tuckute incremental retention **unresolved**	TRIBE-target gain +0.078298, two-sided 95% seed-axis t interval [+0.077198, +0.079398]; mean retained-student movement 0.2151 .	Target uptake and retained-student movement occur within the frozen quality margin; they establish neither attribution to neural target content nor transfer.
Brain-response-specific attribution	**Non-identifying comparator**	The saved text-feature control fails 30/55 frozen measured-axis checks .	Its contrast with TRIBE cannot support attribution to brain-response content.
Biological transfer	**Not demonstrated**	Direct participant-first TRIBE-minus-KD mean −0.000014, 95% interval [−0.000739, +0.000711] .	No participant-general improvement is supported in the tested cohort.
External utility	**Not demonstrated**	Tested routes either lack their prerequisite intervention signal or stop at a precondition .	No brain-specific neural-target payoff claim is licensed; untested routes remain open.

4.1 Controlled regional and naturalistic predictivity

Controlled neural signal is present under both recorded-response assays. On the Tuckute sentence benchmark, the declared middle layers of every trained model family yield positive nuisance-subtracted held-out predictivity, whereas architecture-matched untrained controls are negative across their seeds and evaluated layers . For the primary Qwen2.5-0.5B layer-12 assay, controlled unique R^2 is $+0.036 \pm 0.010$ (mean ± SD across five contiguous folds), and the architecture-matched trained-minus-untrained contrast is +0.050 . The five contiguous stimulus folds are the valid partitions for this fixed five-participant-average target; they support a controlled existence result, not participant-level inference.

The signal also survives the naturalistic assay. For UTS03, GPT-2 and Qwen2.5-0.5B exceed their untrained controls by +0.0207 and +0.0277, respectively, over 11,442 independently selected voxels; every held-out-story block has the same trained-over-untrained direction . Spatially dependent voxels from one participant and story folds with shared training stories are sensitivity dimensions, not independent population replicates; a voxel-resampling interval would pseudo-replicate the biological unit. The supported claim is therefore within-person predictivity under the fixed story, nuisance, reliability, and readout protocol, not a population effect.

Together, the sentence and story results pass the measurement stage without being pooled into one estimate. They show that language training adds linearly accessible information about held-out recorded responses across two stimulus formats and target resolutions. They do not show that the signal can be increased by neural-guided training, that an increase would generalize across people, or that it would improve language behavior.

4.2 Quality-aware distillation headroom

Compression leaves observed alignment headroom in the from-scratch distillation assay. At the fixed Tuckute layers, the cold-start GPT-2 student finishes +0.0181 below GPT-2-medium in controlled brain predictivity and retains 0.37 [0.14, 0.58] of the teacher-to-untrained interval under the recorded bootstrap . The random network and pretrained student supply floor and inherited-alignment references, respectively; the cold-start run asks how much of the teacher-to-floor interval KD transmits without inherited student features. This ordering rules out preservation for free in that cold-start run, but it does not isolate why the gap arose.

Language quality is the main competing explanation. Within the same assay, alignment and log perplexity correlate at Pearson -0.88 across the heterogeneous trained and reference arms . The cold checkpoint also remains substantially worse on held-out language quality than the teacher and warm-start arms, while its absolute alignment estimate changes with the declared PCA rank. Cold and warm arms also differ in initialization and optimizer-step count, so their ordering is not a paired causal contrast . The teacher gap is therefore observed headroom, not an identified effect of the distillation objective.

The cross-family analysis strengthens the need for quality control without turning quality into a cause. Across the full range, Pearson r between alignment and tokenizer-comparable log bits per byte is -0.783, with a six-family cluster-bootstrap 95% interval of $[-0.911, -0.500]$. Models within a family are not independent, and the six families largely separate into roughly two model generations; the family-cluster interval is therefore not six independent cross-generation confirmations. Within the capable-model band ($n = 10$), Pearson r between alignment and log bits per byte is -0.501, with Fisher 95% interval $[-0.859, +0.188]$, which includes zero . After conditioning on log parameter count, the partial r between alignment and log bits per byte remains negative at -0.675, with Fisher 95% interval $[-0.857, -0.344]$; however, parameter count and language quality are highly collinear, and model families lack common quality support for an identified family residual . These tests support quality coupling as a design threat, not a universal law or a causal mechanism.

Controlled predictivity and observed headroom therefore license an intervention test, but only at comparable language quality. The next branch must compare paired interventions at matched checkpoints or operating points; neither the teacher gap nor a quality correlation can substitute for that contrast.

4.3 Recorded-response intervention

Recorded-target training produces positive point estimates against a block-permuted target when the response is averaged across participants, but the valid fold analysis does not establish a pairing effect. For Qwen2.5-0.5B at layer 12, the real-minus-permuted point estimate is +0.00316; the valid analysis averages seeds within each of five shared stimulus folds before computing the t interval $[-0.00227, +0.00858]$. Treating seed-by-fold cells as independent would pseudo-replicate technical repetitions and shared stimuli; removing the dominant fold reduces the mean to +0.0014. This supports neither a reliable fold-level effect nor a zero-effect claim, because the interval remains compatible with a small change.

The averaged-target paired run shows the same distinction between a positive trend and an identified advantage. Its apparent paired gain is +0.0081, with a valid five-fold t interval of [−0.0030, +0.0193]; one fold contributes most of the mean . The valid unit is again the shared stimulus fold after aggregating seed repetitions; treating seeds as independent would pseudo-replicate the training procedure. Moreover, the optimized response is the cellwise five-participant average: it estimates alignment with the shared stimulus-locked target, not the mean effect in a population of participants. The block permutation tests whether the correct sentence–response pairing matters under that averaged target; it does not by itself support attribution to neural target content against a geometry-matched nonbrain target. On the fixed WikiText probe, the real and permuted arms have perplexities 55.4 and 55.3, respectively . Their proximity is a paired quality check showing no gross imbalance, not a declared quality match. Comparisons with ordinary KD remain descriptive unless checkpoints or curves are matched at a common perplexity.

Participant-first evaluation removes that inferential ambiguity and does not support an advantage. After taking the median over 15 paired seed–fold cells within each valid participant, the cohort mean real-minus-permuted effect is +0.00010, with participant t interval [−0.00037, +0.00058] . The shared-fold summary is likewise centered near zero at +0.00026, with five-fold t interval [−0.0004, +0.0009]; the exact participant sign test and leave-one-fold checks do not reject the null . The participant analysis had sensitivity below the apparent averaged-target effect, so its result is not explained by merely replacing a large effect with an obviously uninformative assay. It remains a scoped null for this target, model, objective, cohort, and readout rather than proof that recorded responses can never help training. Crucially, each participant effect is formed only after aggregating that person's seeds and stimulus folds. Seed consistency describes the stochastic training procedure, and the shared-fold analysis checks stimulus concentration; neither creates additional biological units.

Averaging diagnostics explain why the two estimands can diverge without invalidating brain encoding as a measurement. Varying the number and identity of participants in the optimization target does not yield a stable monotone intervention curve once random subsets replace the confounded nested sequence . With the encoder frozen and no intervention, however, averaging responses raises controlled encoding predictivity by improving signal-to-noise and emphasizing the shared response component; individual targets already retain positive signal . Thus averaging neither creates independent biological replicates nor fabricates an absent encoding signal. It changes the target and can strengthen measurement, but it cannot turn an averaged-target training trend into participant-general evidence. The dose diagnostic remains descriptive because its participant subsets overlap and target size changes together with participant identity. The frozen-encoder diagnostic avoids that intervention ambiguity: any score increase there comes from response construction and the changed estimand, not student optimization.

Changing capacity, loss, substrate, or trainable parameter set does not rescue the tested recorded-target branch (Table 4). The table gives only the inferentially relevant disposition. Objectives and schedules are documented in Appendix C, secondary-variant dispositions in Appendix E, and claim-relevant sensitivities and full numerical tables in Appendix F.

The narrow conclusion is that none of the tested recorded-response objectives demonstrates a participant-general advantage attributable to correct stimulus–response pairing. The variants rule out specific explanations—insufficient adapter rank at a fixed weight, one five-ROI loss form, and the tested naturalistic or full-parameter routes—without closing all recorded-target designs. The averaging diagnostic also prevents a broader overcorrection: stronger prediction of an averaged response remains a valid measurement of the shared component, even though it cannot support a claim about mean participant benefit. With controlled signal and headroom established but the recorded-target intervention unsupported, the predicted-target branch must

Table 4: Recorded-target intervention variants. A nonpass is scoped to the named manipulation and valid inference unit.

Variant	Target and unit	Strongest valid read	Disposition
Higher-capacity LoRA	Tuckute participant targets; participant unit	Mean +0.00043, participant t interval $[-0.0005, +0.0013]$.	More adapter capacity at the fixed loss weight does not rescue the participant effect; this is not a test of a stronger manipulation.
Contrastive objective	Tuckute participant targets; participant unit	Mean -0.00019, participant t interval $[-0.0008, +0.0005]$.	Changing the loss does not rescue the effect at five-ROI granularity.
Naturalistic voxel-wise LoRA	LeBel story responses; within-person probes	The required real-over-permuted manipulation check fails in the tested cells .	Non-identifying lever failure, not a naturalistic population null.
Full fine-tuning	LeBel story responses; three participants	Gentle quality-preserving training shows no reliable real-over-permuted gain; aggressive training also incurs forgetting .	The tested full-parameter route does not rescue the effect; no nine-participant inference is implied.

begin again at exact-target measurability before target uptake can matter.

4.4 Exact synthetic-target measurability

The predicted-target branch misses its first biological prerequisite: the exact target fails the frozen practical linear-measurability rule. With train-fold PCA-50 and participant as the inference unit ($n = 9$), the estimator is the participant mean of the aligned TRIBE target's held-out unique R^2 increment beyond the imageability-complete nuisance model. That increment is -0.004494, with Bonferroni family-95% two-sided t interval $[-0.010695, +0.001707]$; the simultaneous upper endpoint is below the frozen $+0.002$ continuation threshold . This result excludes a practically relevant unique linear increment only under the fixed target transform, nuisance model, sentence set, cohort, and readout. It does not exclude total or nuisance-redundant target predictivity, nonlinear access, or another target transform or cohort. The aligned-minus-frozen-row-twin contrast is $+0.001758$, with family-95% interval $[-0.003575, +0.007091]$, so row-specific target advantage remains unresolved .

4.5 Synthetic-response manipulation and attribution status

Target uptake and retained-student movement are supported in the saved WikiText continuity assay, not as fresh Tuckute incremental retention. Using the saved technical training seeds ($n = 6$), the fresh-readout TRIBE-target gain over seed-matched KD is $+0.078298$ target R^2, with two-sided 95% seed-axis t interval $[+0.077198, +0.079398]$. Mean centered relative-Frobenius retained-student movement from KD is 0.2151, and every saved technical seed passes the movement check. The maximum absolute language-quality difference is 0.002579 bits per byte, below the frozen 0.005 margin, and every direct comparison passes . Training parameters are documented in Appendix C, while claim-relevant representation sensitivities appear in Appendix F. In the fresh Tuckute PCA-50 assay, absolute target extractability from TRIBE-trained representations is $+0.040725$, with Bonferroni family-95% interval $[+0.039053, +0.042398]$, whereas the incremental TRIBE-minus-KD retention estimate is -0.000855, with Bonferroni family-95% interval $[-0.002246, +0.000537]$. Thus absolute recovery is stable across the technical seed unit ($n = 6$), but fresh Tuckute incremental retention remains unresolved. Training seeds assess the stability of the saved procedure; they are not biological or model-population replicates. The saved block-permutation arm is only a sensitivity analysis because its permutation duplicates or omits rows, so it cannot supply an exact-null attribution claim .

The passed manipulation checks do not establish attribution to neural target content because the saved nonbrain comparator is non-identifying. The projected text-feature arm fails 30/55 frozen measured-axis checks . Before intervention, mean KD-only target R^2 is 0.595327 versus 0.911687 for TRIBE and text features, respectively, leaving 4.582 times as much mean headroom for TRIBE. Frozen margins also fail for covariance geometry, low-level nuisance predictability, global parameter displacement, and centered-Frobenius retained-student movement. Initial auxiliary-loss and gradient comparability remains unresolved because those values were not retained. Same-teacher redundancy is consistent with the text-feature arm's high baseline fit, but the audit does not identify that or any other mismatch as the cause. The larger TRIBE-target gain therefore remains a valid within-target learnability result and a numerical cross-target difference, not evidence for attribution of the change to neural target content .

4.6 Saved-student biological transfer

After failed or unresolved identification prerequisites, the later biological endpoints describe the saved interventions but cannot establish biological transfer attributable to neural target content. The cleaner direct endpoint compares TRIBE training with ordinary KD separately for each complete participant ($n = 9$). Its participant-mean unique-R^2 difference is -0.000014, with two-sided 95% participant t interval $[-0.000739, +0.000711]$. This near-zero estimate does not support participant-general improvement in the tested cohort, but it neither proves an exactly zero effect nor generalizes beyond the fixed target, model, layer, nuisance design, and sentence/ROI endpoint.

The synthetic-response minus text-derived auxiliary-control contrast is numerically positive but does not support attribution to brain-response content. Its participant mean is $+0.000136$, with two-sided 95% participant t interval $[-0.000505, +0.000777]$; the text-derived auxiliary-control minus ordinary-distillation mean is itself -0.000150 . The positive relative mean therefore does not represent a positive direct synthetic-response effect, and the comparator mismatches in Section 4.4 prevent attribution to brain-response content .

The frozen participant-level composed student-to-target-to-response route also does not establish transfer: every family-95% interval spans zero. Within this route, at the participant inference unit ($n = 9$), the synthetic-response minus ordinary-distillation overlap increment is $+0.000031$, with Bonferroni family-95% interval $[-0.000080, +0.000142]$; absolute overlap and the aligned-minus-frozen-row-twin overlap increment are deferred to Appendix F . Because exact-target measurability fails first and twin specificity remains unresolved, these overlap quantities remain downstream descriptive diagnostics rather than independent evidence of transfer.

The first defensible predictive failure is therefore localized at exact-target practical linear measurability, before student-to-brain overlap. That ordering does not make the measurability result a causal explanation of the near-zero participant transfer: the assay does not test every accessible form of target information, and it does not identify which mechanism produced the later endpoint .

Figure 6 aligns participant-level evidence from the recorded-response intervention, exact synthetic-target measurability assay, and saved-student biological-transfer assay without pooling their estimands or scales. In panel a, grey points are participant medians over paired seed–fold cells for recorded brain-response target minus permuted-response control, and the square is their cohort mean. Panel b uses its own scale: the top row shows the aligned target's increment above nuisance, the bottom row shows aligned target minus frozen row-twin control, and the red line marks only the inherited internal continuation reference. Panel c compares the synthetic-response and ordinary-distillation arms on the same scale as panel a. Black squares and bars show participant means and the stated 95% intervals.

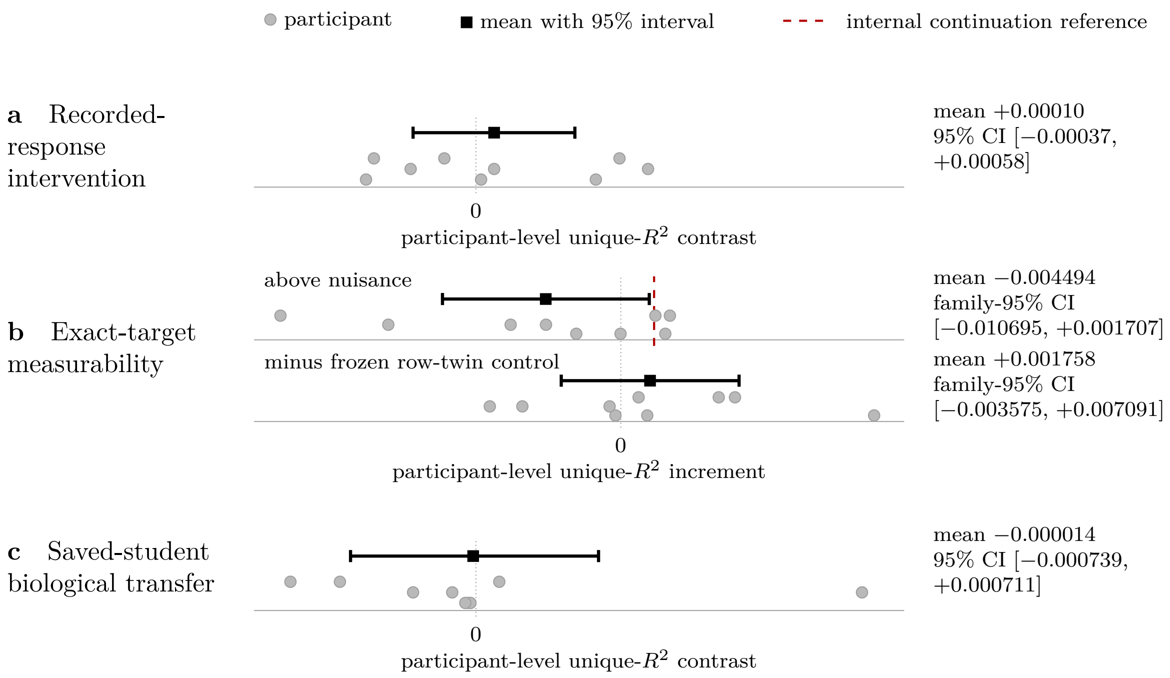

Figure 6: Participant-level evidence at three stages.

4.7 Bounded practical-utility evaluation

No completed route establishes a brain-specific neural-target utility claim, but the routes fail at different prerequisites. The quality-controlled averaged-target test estimates the paired out-of-domain to in-domain perplexity ratio on Pereira text [26] and LeBel text [24], not raw out-of-domain perplexity. Its mediator is the paired real-minus-block-permuted change in in-domain controlled brain alignment. Across the technical training seeds ($n = 8$), that mediator has mean +0.011, median +0.0042, and an 80%-power minimum detectable effect of 0.023; it is therefore unresolved, and no downstream ratio contrast exceeds its measured minimum detectable effect . 5/8 seed contrasts are positive; seed 0 contributes +0.063, whereas the other seven average approximately +0.004. The utility result is consequently bounded to these corpora and this student–teacher pair rather than informative about a successful intervention.

Reduced reproductions, cognitive-target probes, and privileged-information routes provide scoped diagnostics rather than a common utility test. The reduced Negi-style adaptation [9] and single-participant Moussa-style setup [8, 20] do not yield their target benefits; the former is a scoped local lever diagnostic, while the latter is a weak reduced non-reproduction, neither corroboration nor refutation of the original multi-participant result . A content-free shape-matched control preserves the supported generic target-shape regularization finding while removing cognition-specific attribution . The shared-response ceiling and natural-reading privileged-target routes stop at failed instruments or preconditions . Route-specific designs and failure details are deferred to Appendix E.

Unexecuted prospective designs contribute no evidence to this conclusion. The completed evidence therefore supports no brain-specific neural-target utility claim under the tested models, targets, controls, and endpoints, but it does not establish a universal utility null. Controlled neural signal, target uptake, and retained-student movement are observable; the tested chain loses identifying support before participant-general biological transfer and external benefit. Section 5 examines what that ordered failure localizes and how the control design should change.

5 Discussion

5.1 Failure localization across the evidence chain

The evidence chain did not fail at a single common point. Controlled brain predictivity passed in both recorded-response assays . Compression left observed alignment headroom, although language-model quality remained a competing explanation for that headroom . The recorded-target branch then lost support at the intervention stage: none of the valid inference units established a participant-general advantage from correct stimulus–response pairing . In the predicted-target branch, the exact target failed the frozen practical measurability rule, while target specificity against the frozen twin remained unresolved . Target uptake and retained-student movement passed in the saved WikiText continuity assay at acceptable language quality . Fresh Tuckute incremental retention remained unresolved . The saved nonbrain comparator was non-identifying . Participant-general biological transfer was not supported . No tested route subsequently established a brain-specific neural-target utility claim . This pattern separates a measurable signal and a successful manipulation from an identified biological training benefit.

For recorded neural targets, the participant-first result weakens the interpretation that positive trends for an averaged response generalize to people. Response averaging remains useful for measuring a shared, higher-signal component, but it changes the estimand and does not create participant replicates . The intervention variants narrow the failure further without making it objective- or capacity-invariant. Increasing adapter rank at a fixed loss weight did not induce greater effective movement, so it tests added adapter capacity under the same optimization pressure rather than a stronger manipulation. The contrastive result is limited to the tested five-ROI objective, while the voxelwise and full-parameter results close only their tested naturalistic data, loss, and optimization regimes. They weaken those specific rescue explanations, not the broader possibility that another recorded target or induction regime could work .

For predicted neural targets, the exact target did not meet the frozen practical criterion for unique linear predictivity of the intended recorded-response substrate, and its row-specific advantage over the frozen twin remained unresolved . In the saved WikiText continuity assay, target uptake and retained-student movement passed after the temporary head was removed, and the direct comparisons preserved language-model quality, making a simple implementation failure less plausible . Fresh Tuckute incremental retention nevertheless remained unresolved . The saved nonbrain target also differed in baseline predictability, remaining headroom, covariance geometry, nuisance predictability, and induced movement, so the larger target-learning gain could not be attributed to neural target content . The near-zero direct participant endpoint then supplied no participant-general transfer evidence . The first defensible failure is therefore before biological transfer, but neither the measurability result nor the comparator audit explains the later endpoint causally.

Several explanations are consequently weakened, whereas others remain live. The nuisance and untrained-network controls weaken the claim that no controlled neural signal was available to the declared linear readouts . The target uptake, retained-student movement, and quality checks weaken no-op training and gross language-quality damage as explanations for the saved predicted-target result . They do not distinguish insufficient target information from unfavorable target geometry, a mismatch between slow or averaged fMRI targets and the representations being trained, inadequate temporal precision, poorly directed optimization pressure, limited model or data scale, information accessible only through a nonlinear map, or heterogeneous participant effects. These are hypotheses suggested by where support was lost, not findings about why it was lost. The conclusion is therefore a scoped localization over the tested targets, transforms, objectives, models, participants, controls, linear assays, and endpoints, not a universal null for neural supervision.

5.2 Relation to positive prior studies

The local negative result is compatible with positive neural-guided studies because the interventions and estimands are not interchangeable. As Section 2.3 details, the literature spans speech and text models, slow fMRI and temporally resolved recordings, participant-specific and pooled supervision, adaptation and compression, and biological and behavioral endpoints. A high-temporal-resolution target can constrain token- or word-scale representations differently from a sentence-level or hemodynamically blurred target. Personalized training can exploit stable within-person structure that an across-participant claim must treat as heterogeneity. Likewise, a larger retained encoder, a more deeply sampled participant, or a larger training budget can make a target learnable under conditions that need not hold for a fixed smaller student. These differences change the intervention being tested rather than merely changing its implementation details.

The claim licensed by a positive contrast also depends on its comparison and analysis. A gain over a pretrained model answers a different question from an incremental gain over ordinary KD at comparable language-model quality. A gain over a valid, marginal-preserving same-target permutation supports correct stimulus–target pairing but does not distinguish neural target content from other correctly paired, learnable stimulus-derived targets. The defective saved block permutation does not meet that condition and therefore cannot support clean pairing attribution . Repeated training seeds establish procedural stability, whereas participant-level uncertainty is required for a claim about people; folds and neural measurement units answer still different sampling questions. Endpoint choice matters in the same way: improvement on the optimized neural target is target uptake, improvement on newly recorded responses is biological transfer, and improvement on a task is utility. A positive result can be valid for any one of these estimands without satisfying all later stages.

The present study therefore does not refute positive results obtained with different modalities, participant designs, model budgets, baselines, or endpoints. It adds a narrower compatibility judgment: controlled brain predictivity can coexist with an unsupported training advantage, and successful target uptake can coexist with a non-identifying comparator and absent participant-general transfer. Its contribution is to keep those claims separate and to show which controls and inference units are needed when moving from one to the next. Whether a positive prior result would survive every additional stage is an open empirical question, not a conclusion licensed by the present experiments.

5.3 Implications for future brain-guided training

The comparator failure makes target matching a design problem that must be solved before attribution to neural target content. A nonbrain target should match the neural target's accessible stimulus information as closely as the scientific question permits, along with dimensional scale, covariance geometry, nuisance predictability, baseline predictability, remaining headroom, held-out learnability, and the optimization pressure applied to the student. Initial auxiliary losses and gradient scales should be retained so that pressure can be audited rather than inferred after training. Equal output width or a common training schedule is insufficient when one target is already easy for the KD student or induces substantially different movement. This recommendation follows from the observed non-comparability; it does not imply that one fixed matching recipe will identify neural target content in every setting .

The exact-target failure also changes the order of work for predicted neural targets. Before student training, the exact transformed target should be tested on the intended recorded-response substrate using held-out data, the planned nuisance model and readout, an appropriate frozen target control, participant-level inference when the claim concerns people, and a predeclared threshold large enough to matter for the downstream study. The pre-training test should require both practically measurable unique signal beyond nuisance and adjudicated target specificity relative to that control. In the present assay, the unique increment failed the frozen practical rule and aligned-versus-twin specificity remained unresolved . Failure at either part should trigger a target, transform, modality, control, or assay redesign before expensive optimization. Passing both would not guarantee transfer, but it would establish under that assay a specific measurable signal that the later biological transfer test could in principle preserve. The observed failure supports this ordering for diagnostic efficiency, not a universal claim that linear measurability is necessary for every possible nonlinear intervention.

For synthetic brain-response targets, the decision rule is conjunctive: exact-target measurability must pass; target uptake and retained-student movement must then pass at acceptable language quality; brain-response-specific attribution must be established against a matched text-derived auxiliary target, including matched trained-model learnability and movement; and biological transfer must be supported on an independent recorded endpoint at participant level . After target uptake and retained-student movement pass, evaluation must move away from the optimized target. The direct target-trained-versus-KD contrast should be measured on independently recorded responses, with technical seeds and stimulus folds aggregated within each participant before population inference. A valid matched comparator is additionally required to attribute any transfer to neural target content rather than generic auxiliary supervision. Reporting target uptake, retained-student movement, language quality, and participant-level transfer as separate outcomes makes the failure location visible: a manipulation can be reliable even when its biological endpoint is not. Only after every prerequisite passes does a downstream utility test estimate the payoff of an identified biological intervention .

The failure stage should determine whether a study continues, redesigns, or stops. When target uptake and retained-student movement pass but participant-level transfer remains unresolved, strengthening or repeating the transfer test is justified only if added participants, a better independent endpoint, or a more informative assay could change the decision. When a practically meaningful effect is excluded or an upstream stage fails, the warranted choices are to stop or redesign the target, control, or assay rather than accumulate more target-fitting evidence. Utility testing should wait in either case. This decision rule avoids spending downstream compute to interpret an upstream failure and avoids treating every nonpass as the same kind of null. It is a recommendation derived from the diagnostic pattern in this study, not an experimentally proven universal recipe for brain-guided training.

6 Limitations

6.1 Population, data, and measurement scope

The biological coverage is limited to English-language fMRI responses from the studied cohorts. The sentence experiments summarize responses within fixed left-hemisphere language ROIs, while the naturalistic assay uses reliability-selected voxels from a few deeply sampled story listeners. Some training targets average responses across participants to emphasize a shared stimulus-locked component; the participant-first analyses instead retain one effect per person. The conclusion therefore concerns these isolated-sentence and spoken-story response summaries, not language processing across the brain or population at large .

The experiments do not cover neural modalities with finer temporal resolution, populations that differ in language background or clinical and developmental status, non-English or nonlinguistic stimuli, or a broader range of reading, listening, and interactive contexts. They also do not test whether information unavailable to the declared linear ridge readouts is accessible through nonlinear maps or at different temporal scales. Because modality, temporal precision, population, and stimulus structure can change both the target and the intervention, the local failures do not extend to those untested regimes .

Identification is conditional on the declared nuisance sets, controls, and dependence structure.

6.2 Identification and inference limitations

Unique predictivity is only beyond the included nuisance features, not beyond all stimulus-derived information, and the trained-versus-untrained contrast establishes learned predictive access rather than a shared mechanism . The saved nonbrain comparator mismatches target predictability, geometry, learnability, and induced movement, so its cross-target contrast cannot identify neural target content . Exact-target measurability is further restricted to the PCA-50 linear assay, and specificity against the frozen twin remains unresolved . Shared folds, stories, voxels, and training seeds are dependent measurements rather than population replicates; consequently, participant-general claims remain limited to analyses that aggregate them within person.

The participant intervals disfavor effects at the declared continuation scales in the tested cohorts, but they do not establish exact zero. For recorded-target pairing, the participant interval is $[-0.00037, +0.00058]$, far below the earlier averaged-response trend while still including zero . For the predicted-target comparison, the preregistered relative upper bound is $+0.000653$, below the internal $+0.002$ continuation scale, although the comparator prevents attribution to neural target content; the direct target-trained-versus-KD interval is $[-0.000739, +0.000711]$. The exact-target unique-increment interval is $[-0.010695, +0.001707]$, whose simultaneous upper endpoint is also below that inherited scale . These bounds disfavor participant-average intervention effects and target-measurability increments of the prespecified size under the tested assays, while remaining compatible with smaller effects, mixed participant responses, and effects in other cohorts.

The intervention search was diagnostic rather than exhaustive.

6.3 Intervention and utility scope

It did not jointly explore a broad range of student scales and architectures, objective families, loss weights, schedules, data volumes, target constructions, or participant-personalization strategies. Increasing adapter capacity at a fixed loss weight did not create a stronger effective manipulation, and the contrastive, voxelwise, and full-parameter variants close only their tested parameterizations and data regimes . The conclusion therefore localizes failure within the executed search rather than claiming that a different scale, target, personalized objective, or optimization schedule cannot succeed.

Practical utility was also sampled through a narrow set of endpoints. Perplexity served both as a language-quality control and as the primary out-of-domain utility measure, so the experiments do not exhaust task accuracy, calibration, robustness, sample efficiency, or deployment-specific value . The Negi-style experiment is a scoped lever diagnostic because protocol and substrate differences prevent a faithful reproduction of the source study [9]. The reduced Moussa-style experiment is a weak single-participant non-reproduction of only the local setup [8, 20]. Neither result refutes its source study. The gaze route stops before its planned intervention, and the shared-response route stops at a failed instrument . Accordingly, the evidence licenses no brain-specific neural-target utility claim for the completed routes, but it does not establish that every practical endpoint would be unchanged after a successfully identified biological intervention.

7 Conclusion

For the tested English fMRI systems, controlled brain predictivity and retained-student manipulation are observable, but neither tested branch establishes a participant-general, brain-specific training benefit, and no completed route licenses a brain-specific neural-target utility claim .

In the recorded-target branch, trained middle-layer representations show controlled brain predictivity , and compression leaves quality-coupled headroom whose source remains unresolved , yet participant-general pairing is unsupported across the tested objectives and variants . In the predicted-target branch, saved WikiText target uptake and retained-student movement pass at acceptable language quality , but those manipulation passes do not overcome failed exact-target practical linear measurability , a non-identifying comparator , or near-zero participant-level biological transfer .

The durable lesson is that a brain score becomes evidence for a training signal only through an ordered chain of separately supported claims. Controlled brain predictivity and quality-aware headroom must be followed, where applicable, by exact-target measurability, target uptake and retained-student movement at acceptable language quality, brain-response-specific attribution against the route-appropriate control, participant-level transfer to independently recorded responses, and an endpoint appropriate to the utility claim. A pairing permutation and a matched nonbrain target answer different attribution questions, while seeds, folds, and neural coordinates cannot replace participants as the inference unit for claims about people. This ordering makes clear what an intervention achieved, where identifying support stopped, and what evidence would be required before a measured brain score could be treated as a useful training signal.

A Evidence Provenance and Analysis Disposition

The experiment record is the authority for what was run and what an artifact supports. The ledger below distinguishes scientific results from engineering checks, stopped designs, and unused identifiers. Its last column names the main-text sections that use each record: F = Experimental Framework, R = Results, D = Discussion, L = Limitations, and C = Conclusion. "None" means that the record is retained for provenance but does not support a current main-text claim.

Table 5: Disposition, final use, and main-text support for E001–E030.

Record	Disposition	Settled verdict and permitted use	Main text
E001	Engineering only	Synthetic smoke tests validate the apparatus; no real-neural scientific result was produced.	F
E002	Complete	Controlled Tuckute ROI measurability passes under contiguous folds, nuisance subtraction, and an untrained-network control.	F, R, D, L, C
E003	Complete; narrowed	Ordinary KD leaves quality-aware headroom, but its source remains unresolved because retention is entangled with language quality and objective-specific shedding is unidentified.	F, R, D, C
E004	Complete; reanalysed	The recorded-target intervention does not demonstrate a gain when uncertainty is computed over held-out folds rather than seed–fold cells.	F, R
E005	Complete; reanalysed	The averaged-target intervention shows a small trend; the original excludes-zero claim is superseded.	F, R, D
E006	Complete; corrected	Trained features outperform architecture-matched untrained features for fixed UTS03 and five held-out story blocks; voxel and population inference are not licensed.	F, R, D, L, C
E007	Intentionally unused	No E record or scientific result. The invalid E006-derived minimum-detectable-effect claim was never promoted.	None
E008	Complete	The recorded-target intervention does not demonstrate a participant-general gain across the nine complete Tuckute participants.	F, R, D, L, C

Continued on next page

Record	Disposition	Settled verdict and permitted use	Main text
E009	Complete	The practical-utility route is a bounded null under an unreliable alignment mediator, not evidence that alignment can never be useful.	R, D, L, C
E010	Complete, including the corrected random-subset analysis	Random participant subsets do not identify a count-dose response; the earlier nested-subset monotone reading is superseded.	R, D
E011	Complete	More LoRA capacity and training duration did not strengthen the manipulation, but this was not a stronger neural intervention.	R, D, L, C
E012	Deferred; unrun	Three participants were insufficient for the proposed inference, so no experimental evidence was produced.	None
E013	Tested branches complete; expansion unbuilt	The contrastive ROI branch does not demonstrate a gain and the voxelwise lever failed to take hold; the larger expansion was not executed.	R, D, L, C
E014	Complete	Participant averaging legitimately improves target signal-to-noise ratio while changing the estimand; it is not a universal artificial lift.	R, D
E015	Complete; corrected	Quality and alignment are strongly coupled over the full model range, while the capable-band relation and architecture residual remain uncertain.	F, R, D, C
E016	Complete; independently reviewed	Dense-target uptake and retained-student movement pass, but the dimension-matched comparator cannot support attribution to neural target content and averaged-response transfer is negative.	F, R, D, C
E017	Complete; narrow probe	A three-participant full-fine-tuning existence probe finds no specific gain and supports no population inference.	F, R, D, L, C
E018	Intentionally unused	No E record and no scientific content.	None
E019	Complete; scoped diagnostic	The local faithful-as-feasible lever test shows no gain; it is neither a reproduction nor a refutation of the source study.	R, L
E020	Complete; failed instrument	The implemented controlled residual is bounded, but estimator reliability fails, so the full stimulus-predictable ceiling remains open.	R, L
E021	Complete; corrected	A phase-randomized control removes the apparent cognitive advantage; the final finding is shape-specific, not cognition-specific.	R
E022	Complete; reduced pilot	The single-participant reduced speech experiment does not reproduce the gain and does not refute the source study.	R, L
E023	Design plus analysis-only criterion	The saturated slope/knee comparison was not cleared and no intervention training was run.	None
E024	Preconditions complete; intervention stopped	Gaze informativeness and split-validity criteria fail, so training stops; the electroencephalography (EEG) route remains untested.	R, L, C
E025	Complete; independently audited	Manipulation checks and quality criteria pass, but direct TRIBE-minus-KD transfer is near zero and its upper bound misses the internal +0.002 continuation rule; this does not establish attribution to brain-response content.	F, R, D, L, C
E026	Complete; independently audited	Thirty of 55 frozen comparability checks fail and one is unresolved; the saved `textfeat` comparator cannot support attribution to neural target content.	F, R, D, L, C

Continued on next page

Record	Disposition	Settled verdict and permitted use	Main text
E027	Unexecuted; not authorized	No E record or experiment. The prospective matched-target run was closed when E025 missed its continuation rule.	None
E028	DESIGN HOLD; engineering only	Outcome-blind synthetic executable checks pass, but no endpoint data were opened and no training was authorized; this is not evidence.	None
E029	DESIGN HOLD; validator only	The deterministic algebra validator is a design artifact, not a scientific result.	None
E030	Complete; recomputed and author-confirmed	The fixed PCA-50 target fails the inherited +0.002 rule for exact-target practical linear measurability; aligned-minus-twin specificity is unresolved. This localizes the first observed failure but does not explain E025 causally.	F, R, D, L, C

A.1 Consequential corrections

Nine corrections govern how earlier records may be used. First, E003 supports quality-aware headroom whose source remains unresolved, not a claim that KD uniquely destroys or preserves alignment. Second, E004 and E005 seed–fold cells share evaluation data: fold-unit inference supersedes the 15-cell bootstrap, and E008 supplies the participant-unit test. Third, E006 voxel-bootstrap intervals and the derived E007 power claim are retracted; the retained statement is conditional on UTS03 and five held-out story blocks. Fourth, the corrected E010 random-subset analysis supersedes E010's nested monotone dose-response reading; E014 instead establishes the defensible signal-to-noise and estimand effect of averaging. Fifth, E015 replaces the initial narrow-family, best-layer correlation with a tokenizer-independent 22-model, six-family mid-layer analysis; full-range coupling survives, while the capable-band relation remains uncertain. Sixth, E021's phase-randomized control supersedes the earlier cognition-specific reading. Seventh, the Tuckute value 0.491 is a correlation ceiling and cannot divide variance-scale unique R^2; the raw contrasts and verdicts are unchanged. Eighth, E016's block permutation is defective and `textfeat` is dimension-matched only; E025 supplies the clean near-zero direct participant contrast, and E026 shows that the relative comparator cannot support attribution to neural target content. Ninth, E030 localizes failure at exact-target practical linear measurability, while the aligned-minus-twin contrast remains unresolved; it does not establish why E025 failed.

A.2 Retained load-bearing artifacts

Table 6 is a verified subset of retained provenance, not an exhaustive list of every central result output. It reproduces only path–hash pairs explicitly bound together by an owning E record. It excludes smoke outputs, intermediate caches, prospective-design artifacts, and any hash whose record does not bind it to an exact path.

Table 6: Stable retained artifact paths and recorded SHA-256 values.

Record	Retained path and SHA-256
E002	`outputs/E002_tuckute_feasibility.json` SHA-256: `ce45eb1786514a168e81248c426c472077a44245d8863ee5d0bbf6e4aacab44d`

Continued on next page

Record	Retained path and SHA-256
E003	outputs/E003_cold.json SHA-256: a30dceb63832a2fce838907cc9c782b4f6e8297c2a0136f94b5aacc9056edd6e
	outputs/E003_warm.json SHA-256: 6e75770387b0a84866060049d8fb8d3aecda898a95b2ce159ceffc226a5bf387
	outputs/E003_perplexity.json SHA-256: 2385e67ec61b7c64276f01e7b8e717e1d54cb1c1b23023715922e685cab5f97b
	outputs/E003_dissociation.json SHA-256: b7658aa0d1acf92bd62897aadec663aeda7a662abfa58890ca63eab6db5e5d38
E004	outputs/E004_brain_lever_Qwen.json SHA-256: 5f5582a651ecd86c25691fe24b6827d49a258c33e95127f630c522563d8dc6d4
E005	outputs/E005_Qwen.json SHA-256: e9b7be5b739370ed224de1bd8c6d4fa34d33881db308d3508e1edba66ddfcf6e
	outputs/E005_lambda_sweep_avg_Qwen_analysis.json SHA-256: 85013e6ff961aa88bd40e9e16dd19aaab3c89656f0dac8cd74c64560469f7938
E006	outputs/E006_lebel_gpt2.json SHA-256: ea424b7d2f8a32dd289e8ef8066c8609749f4edc1b4cdd132373a78bd6b6830b
	outputs/E006_lebel_Qwen.json SHA-256: 09b77b8801e937d8232ce7e7f16435e490f0b9300c78ce4ede23d00bf07d2581
E008	outputs/E008_g0.json SHA-256: fa2d18629e3818412a493539438d211c60cb8052de13286dd632983642d056cd
	outputs/E008_g1.json SHA-256: 0a60a95009f8bdb2c9aca1ba8f17f9e88f2f1be584a7db13bd35f6c9b51e6b5a
	outputs/E008_g2.json SHA-256: e51b58d2f4b19fa624ed7b46dd0e4bfe35d5fb5b57104d78aa06a269830aaecf
	outputs/E008_g3.json SHA-256: 70903174700a20b5eaf4b44f685a6e14317daff3f35c14b28d1d4ac645ac5803
E009	outputs/E009_a3_powered.json SHA-256: 1820c097a1a30d3a23bb0956221fbfb5b33c6dc67557e108ac6b79dc56c46afb
E010	outputs/E010_averaging_doseresponse.json SHA-256: 79312a649c3021df1fdc94aa46f3fcaf6c9b57fa30e59bd4fcd2e28db87ba5d0
	outputs/E010b_random_subsets.json SHA-256: 276d19bbf013f4f90d9ccbbd02151926a67fdc5af08dc5787902420d754f1d17
E011	outputs/E011_g0.json SHA-256: 299ca57c7733932d9272a44d06fe3cb6976ec535a887b5ad27072002426ea9e2
	outputs/E011_g1.json SHA-256: f3963b67cc01e0fae3807de354b613d4e6a3ca6ff8c6e55e1eed6b7c8b719bd3
	outputs/E011_g2.json SHA-256: 136449382dc627d5007df84b248b3c63d8b9bd4e2d1164c7399afbb9b867b4ce
	outputs/E011_g3.json SHA-256: 4da6639f3943c931beab6b7189fd6a03248161c846180a225e0e9c17af5b0745
E013	outputs/E013b_g0.json SHA-256: a98346f08ad41103d1cc7b9c28715dd810927c3fa09c63bf4d1db858ece19976
	outputs/E013b_g1.json SHA-256: e472ee523b6af62701472bca554f79eed64f319944549349b01d6b54d3b43346
	outputs/E013b_g2.json SHA-256: f9df4143b2b1cce4e871778f30a0b4f2bb48f561f4847693be0b5a4bd2c03361
	outputs/E013b_g3.json SHA-256: ae741c2c4e948356841e45b9053686823071365f8cfb9e362018b66213126cad
	outputs/E013v2_check.json SHA-256: a7ea8adbda9312c55192015f1b704948d13774fc2105937ac6b58d159c6e99de
	outputs/E013v2_lam3.json SHA-256: 35869107e166aade0097ef2e337d133a2ae98b27d7f494c1dc3a7b4b09c6a926
	outputs/E013v2_lam6.json SHA-256: dcb0ab4a31f21da71cf2aea87bdfdada8a1eb530d2955fa8923eccde57e3421a
E014	outputs/E014_averaging_encoding.json SHA-256: b8f25449e313b17119b908a98e2c3647ca8749592ab9024b215d233034a22259
E015	outputs/E015_expand/E015_expand_merged.json SHA-256: 0aa4f149c1ae471f3b008ff77707570e2df9c4e3a5df7e61c189aa87f438abad
	outputs/E015_expand/panel_followup.json SHA-256: 6c5d9738e08028636a5bbbf21dfad1acde2da5756edeb3c628dd3f8664a6eefa

Continued on next page

Record	Retained path and SHA-256
E016	outputs/E016_tribe/phase3/phase3_combined_tribe_gpt2_n95999_s0-5_lam0.1.json SHA-256: 9fd5a657bde1d95cfad93a85a3e6513514c33b45d21d7245ef0a0336f7ee9d98
	outputs/E016_tribe/phase3/phase3_combined_textfeat_gpt2_n95999_s0-5_lam0.1.json SHA-256: 793af7beaab54aff1267e44ef16bdffa5b4f5854d8fc99bcfc99a2bbec0d382d
	outputs/E016_tribe/phase3/phase3_combined_tribe_vs_textfeat_s0-5_comparison.interpret_audit.json SHA-256: 2716234438f2b0a0519fab9d16bceebaffea2e1c6fc93e9cd6f90a6819df2f5e
	outputs/E016_tribe/phase3/phase3_rerun_tribe_gpt2_n95999_s0-2_lam0.1_save.tuckute_alignment.json SHA-256: f7ea1607f48bb6b91c9e9f4087ac691aca527ac4c46ddfecb758fd90554827f7
	outputs/E016_tribe/phase3/phase3_extra_tribe_gpt2_n95999_s3-5_lam0.1.tuckute_alignment.json SHA-256: cf5e7805bc5bd5fec4b6bd5df0d652b1b28368741c874d686c19ffa0c37096ac
	outputs/E016_tribe/phase3/phase3_combined_textfeat_gpt2_n95999_s0-5_lam0.1.tuckute_alignment.json SHA-256: fa94779f2b0a06c748ef9016b9494720d82c9b345ea9fb577bc4bde6ee898024
	outputs/E016_tribe/phase3/phase3_combined_tribe_vs_textfeat_s0-5_tuckute_alignment.interpret_recompute.json SHA-256: 9fe1a4253bc39b1b75ce5d25184332825706b95518a493532af3a276d10f10eb
E017	outputs/E017_matched_ppl_control/fullft_gentle_3subj.json SHA-256: 8f426e7b012f8461208bfb6a0e3e0bf3981eacf2feef91fb95f40638e5ca4e38
E019	outputs/E019_negi/results.json SHA-256: 983805956f37d8e17d81d3172113d46bdfe174c6ccce581ce5d1ae040d165758
	outputs/E019_negi/probe_strong.json SHA-256: 2fed0402fcce0067a81d81330289829c6fd1d6c5128dfb30cccda14f9787664b
	outputs/E019_negi/eval_posctrl.json SHA-256: c7a9a5f5c07b3802842c73dbde891f6e9185751f3898522dd13c8347c45c214f
E020	outputs/E020_eys/eys_story11_results.json SHA-256: d16d6f94f2f8f389db3178262bb399c93a8f973321684e1246bf1009f063acab
	outputs/E020_eys/eys_diagnose.json SHA-256: e8fab13e0fed3840a94fb9605d2ca4868ddc6e78d6cc975cede6f7f03b8579b0
	outputs/E020_eys/eys_diagnose2.json SHA-256: 84d268dc4a6211e7775cc5bbeca84e5d64eb52fbd8af963799011014eacb1879
E021	outputs/e021/arms_v5_results.json SHA-256: 95c05b8d259d82fb83c84d54c595176198a9ba9cfe50a70941c2cddd24d69b5b
	outputs/e021/arms_v5_shard_all.json SHA-256: b53bf1c88de36b397a9be753587e0e3846fde107c648eb496250850a1b7314e1
E022	outputs/e022/pilot_results.json SHA-256: a95a1e23be864322fce4f26ad215927b8fb74739ae35156c0da0091a0eccd24c
E024	outputs/e024/zuco_nr_reliability.json SHA-256: 34f412c1d40177510e2369296a223c00267da1aa3af940514ece47c3e5dfc4e2
	outputs/e024/positive_control.json SHA-256: acea6d09191cbd1e31062dd4c3ed1963de5ff95387f87f4a8d26b836edaa8e1a
	outputs/e024/gaze_richness_probe_NR.json SHA-256: b8eb0f32bd16e7eda77f957f23ad329c881ce2f550e9e5f4de0b48bef4ae606e
	outputs/e024/gaze_richness_probe_TSR.json SHA-256: 5d23870ad33f24a52044983446b347e51a9b6ee437c1e3659eaa633e8cc694a8
E025	outputs/E025/extraction.json SHA-256: b49c2795c55961cae161084afa1d3409dd662b893f9a40fa6a4e9373bc2c84c5
	outputs/E025/analysis.json SHA-256: 9a43410bfafced8130d286faa8c011c6e2097707979d476b5ce01d9ec44f106c
E026	outputs/E026/e026_audit.json SHA-256: dfac7498d89d488483f129ebd30f36013d98778a8304f0d3d97d9e0a8c1932ad
	outputs/E026/e026_independent_result_audit.json SHA-256: b2dca54c87efeca60c78a0de8287b4571991dcf3e653ce17dacc60583d2fda9f
E030	outputs/E030/tribe_tuckute_condition-b_n1000_full.npz SHA-256: a1fb8667e5c27a7e9bde35cb5f3a4f24622ad13c3b79e91a884c119ef3b65233
	outputs/E030/raw_scores.json SHA-256: cebea58cd7f7d82908b26a25714a1d256ab674a7b08d9a18d94835b0f7fcb6e4
	outputs/E030/analysis.json SHA-256: 43a4f3b01641072345149e651cdd0b1ae0c903bb74366db2006f31b89d8dd818

The owning records now bind the E025 extraction and analysis paths and the E030 raw-score and analysis paths used by the numerical appendices. E025's raw fold-grid hash and several E030 bundle or manifest hashes remain unbound to exact filenames, so the 59 verified pairs above remain a curated subset rather than an exhaustive artifact registry. E028 and E029 artifacts are omitted because they document outcome-blind design or engineering checks rather than experimental evidence.

B Data, Responses, Targets, and Controls

B.1 Data inclusion and response construction

The Tuckute resource was acquired during silent sentence reading with 3-T fMRI, a 2.0-second repetition time (TR), and 2-mm isotropic voxels [23]. The controlled substrate is condition B: 1,000 English sentences ordered by `item_id`, with released blood-oxygen-level-dependent (BOLD) responses indexed by participant, sentence, and five left-hemisphere language ROIs (`AntTemp`, `IFG`, `IFGorb`, `MFG`, and `PostTemp`). This repository does not reconstruct Tuckute voxel time series or refit the functional localizers. It inherits the released `response_target`, for which each voxel was z-scored within scanning session before voxel responses were averaged inside an ROI; transformations here begin with condition selection, deterministic row ordering, and participant aggregation. The controlled-measurement and first intervention experiments use a cellwise participant average over the original training unique identifiers (UIDs) 848, 853, 865, 875, and 876. The group operation skips a missing contributor, but the resulting sentence-by-ROI matrix must be complete before it is scored. Participant-first analyses instead retain one response matrix per person. UID 853 is excluded because 60 entries are missing and the five-ROI grid is incomplete; the valid units are UIDs 797, 837, 841, 848, 856, 865, 875, 876, and 880. The inherited training-participant subgroup is therefore {848, 865, 875, 876}, and the held-out-participant subgroup is {797, 837, 841, 856, 880}. All analyses retain the same 1,000 rows and fixed ROIs; neither participant nor sentence inclusion is selected from an intervention outcome.

The participant average is formed before ridge fitting and before the nonlinear held-out R^2 calculation. It estimates predictivity of the shared response of that fixed group, not the expected response of a randomly sampled person. Participant-first estimators instead summarize folds and technical seeds within a person using the declared estimator and use the nine people as the biological inference units. These targets can therefore differ in signal-to-noise ratio and in scientific meaning even when they use identical sentences and features. The Tuckute value $r_{\mathrm{NC}} = 0.491$ is a correlation-scale reliability reference for the five-ROI average; it is not a denominator for variance-scale unique R^2.

The naturalistic assay uses participant UTS03 from the LeBel story-listening resource, acquired during auditory narratives and released as preprocessed storywise matrices sampled every 2.0045 seconds [24]. Each HDF5 `data` matrix contains one row per scanner sample and 95,556 cortical-voxel columns; this work inherits that released imaging representation rather than re-running reconstruction, motion correction, or cortical registration. Transformations performed here begin after loading those matrices: the response columns are z-scored within story, aligned to the transformed stimulus rows, and filtered by an independent reliability screen. The exact 20-story order is retained in `outputs/E006_lebel_gpt2.json`, and the frozen contiguous splitter forms these held-out folds:

- Fold one: `adollshouse`, `adventuresinsayingyes`, `afatherscover`, `againstthewind`.
- Fold two: `alternateithicatom`, `avatar`, `backsideofthestorm`, `becomingindian`.
- Fold three: `beneaththemushroomcloud`, `birthofanation`, `bluehope`, `breakingupintheageofgoogle`.
- Fold four: `buck`, `catfishingstrangerstofindmyself`,

cautioneating, christmas1940.

- Fold five: cocoonoflove, comingofageondeathrow, exorcism, eyespy.

The fold membership follows the recorded story order and the executable's contiguous four-story partition; wheretheressmoke is excluded from this pool. Voxel inclusion is determined independently from its released individual_repeats array: ten repetitions, 291 scanner samples, and 95,556 voxels. Split-half reliability is Spearman–Brown corrected, and only voxels above 0.5 are retained, leaving 11,442 voxels. The repeated story is used for reliability selection, not for fitting or evaluating the language-model readout. This construction supports a conditional within-UTS03 result; many voxels do not create additional biological replicates.

B.2 Stimulus and nuisance features

For the isolated-sentence assays, a contextual feature is the attention-mask mean of the declared language-model layer. The base nuisance design contains sentence token length, normalized item position, and a mean noncontextual input-embedding vector. The participant-transfer implementation freezes the static block to GPT-2-medium so that every student arm receives the same nuisance representation. The exact-substrate primary design adds the released imageability covariate; the byte-identical base design without imageability is retained as a continuity sensitivity. GPT-2XL and probabilistic context-free grammar (PCFG) surprisal form a nonterminal stress test, not the primary nuisance set, because model-derived surprisal overlaps the contextual construct whose incremental value is being tested.

Each outer split is one of five contiguous 200-sentence blocks. Within every split, stimulus-domain scaling, PCA, nuisance regression, ridge selection, and fresh readouts are fitted on the other 800 sentences and then applied unchanged to the held-out block. The exact-substrate target is the explicit exception: it inherits the externally frozen E016 target standardization rather than refitting those moments on each 800-sentence outer-training set; any subsequent fold-specific target PCA is still fitted on outer-training rows only. Contextual and static blocks are reduced separately to rank 50, while scalar nuisances remain unprojected. Using equal ranks prevents a wider contextual matrix from receiving more readout capacity by construction. Sentence order is never randomized across train and test, and ranks, nuisance tiers, and participant subsets are fixed before held-out scoring. E002 reports the peak over its fixed layer grid, whereas the intervention assays lock their primary and sensitivity layers before their own outcomes.

For LeBel, the transformations performed here use TextGrid word and phone tiers to define word identities, word onset times, and phoneme events. Word duration is approximated by clipped successive word-onset differences from 0 to 3 seconds, with a 0.3-second terminal-word proxy. Contextual word features are extracted in overlapping 256-token windows with stride 128, assigning each word the hidden state at its final subtoken under the maximum available left context. The low-level nuisance block contains word rate, phoneme rate, mean word duration, word length, and log word frequency; a 985-dimensional english1000 lexical-semantic vector supplies the static block. Word length, word frequency, and word duration are Lanczos-resampled with window 3 to the 2.0045-second fMRI grid, as is the english1000 stream; word and phoneme rates are instead direct counts in TR-centered histogram windows. The language-model word features are Lanczos-resampled separately with the same window. After these stream-specific constructions, the language-model, low-level, and static streams share the fixed slice [10:-5], within-story z-scoring, and finite-impulse-response copies at delays one through four TRs.

The inherited BOLD matrix is z-scored voxelwise within story but receives no finite-impulse-response expansion; feature and response rows are then truncated to their common aligned length. The contextual and static blocks receive train-fold PCA to rank 100, while the low-dimensional nuisance variables remain unprojected. Within each fold, the PCA bases are learned from training stories, and a predictor `StandardScaler` supplies additional train-fitted z-scoring to each assembled design; the response mean, shared ridge hyperparameter, and fitted readout are also learned from training stories. The earlier within-story z-scoring, trimming, and finite-impulse-response expansion are fixed preprocessing. Complete stories, rather than neighboring TRs, are held out, so temporally autocorrelated samples cannot cross the evaluation boundary. The same story folds, nuisance arrays, temporal transforms, and response rows are used for trained and architecture-matched untrained networks.

B.3 Synthetic brain-response and text-derived control targets

TRIBE v2 maps stimulus events to predicted fMRI activity on a standardized 20,484-location cortical surface [27]. It is therefore a predicted neural target, not a neural recording. The continuity target builder loads the `facebook/tribev2` checkpoint, sets `data.text_feature.model_name` to `unsloth/Llama-3.2-3B`, and sets `data.features_to_use` to `[text]` through the operative runtime override. Its `synthetic-text` configuration assigns each token a 0.35-second event followed by a 0.05-second gap, extends timelines shorter than 1.0 second, uses a 1.0-second TR, retains all 20,484 fsaverage5 vertices, and processes 32 items per batch. Each sentence has a private timeline, and all retained one-second predictions on that timeline are averaged coordinatewise to obtain one target vector. The resulting caches contain 95,999 training sentences and 1,999 untouched held-out sentences. The exact-substrate cache applies the same checkpoint, override, vertex order, and event-timing configuration to Tuckute item IDs 1–1,000, bound to generator indices 0–999 by `item_id=item_index+1`. Training-standardized coordinates use

$$\widetilde{Y}_j = \frac{Y_j - \mu_{j,\text{train}}}{\sigma_{j,\text{train}} + 10^{-6}},$$

with moments computed from float32 targets without a dtype override, stored as float32, and applied in float32. The student layer-6 sentence representation is connected to all standardized target coordinates by a jointly trained linear head; no PCA is used in this auxiliary training loss.

The `textfeat` arm is a nonbrain target built from mean-pooled GPT-2-medium layer-12 features after tokenization to at most 64 positions. Its projection implementation draws one float32 matrix with `numpy.random.default_rng(0).standard_normal((1024,20484))`, multiplies it by $1/\sqrt{1024} = 0.03125$, and right-multiplies the float32 1,024-dimensional source matrix. The resulting 20,484 columns then pass through the same standardization, head, corpus, optimizer, and six-seed training path. This matches row identity, output width, and much of the training machinery. It does not demonstrate matched accessible stimulus information: the projected source has centered algebraic rank at most 1,024, and equal marginal scale does not equalize covariance spectrum, effective rank, baseline predictability, or remaining headroom.

The comparator audit examines the two saved targets rather than assuming that dimension matching is sufficient for attribution to neural target content. On two disjoint deterministic samples of 4,096 training rows it reconstructs the training standardization and compares centered covariance spectra, stable rank, cumulative variance mass, within-document order structure, low-level and static-feature predictability, KD-baseline target recovery, and remaining headroom. Across the six trained seeds it also compares parameter displacement and held-out layer-6 representation movement from byte-identical KD checkpoints. The observed mismatches in rank, covariance geometry, KD-baseline predictability, and remaining headroom show that the saved comparison cannot support attribution to neural target content. Initial-gradient comparability remains unknown because the initial gradients were not retained; the audit does not classify those gradients as unequal. Thirty of 55 frozen measured-axis checks fail and one is unresolved, while `textfeat` remains usable as a descriptive nonbrain arm.

Two different row controls must not be conflated. The saved block-permutation sensitivity permutes contiguous target blocks, but it is not a derangement: depending on seed, 0–30% of rows remain correctly paired, and unequal blocks duplicate one row and omit one row in five of six seeds. It is retained only as a defective-control sensitivity. The frozen Sattolo twin instead applies one hash-keyed derangement separately inside each 200-item outer block before target values or participant scores are read. That twin has no fixed points and preserves the exact row multiset within each block. For each outer fold, aligned TRIBE rows determine one train-only target PCA-50 basis, and the same basis is applied to the twin; nuisance models, representation PCA-50 transforms, ridge choices, and composed readouts are also fitted only on outer-training rows. The twin therefore controls row identity at matched target capacity, but one frozen twin cannot establish population-level invariance or a causal neural mechanism.

C Training Interventions and Reproduction

This appendix specifies what was optimized, which parameters could receive each gradient, and what survived as the evaluated student. The common distinction is between a permanent language-model path and, when present, a temporary target head attached to one middle state. Only the permanent student is retained. All reported response- or target-predictivity scores are obtained later with a fresh readout.

C.1 Ordinary distillation and language-quality control

Let $z_T(x)_t$ and $z_S(x)_t$ be teacher and student logits at a non-padding causal-language-model position t, and let

$$p_T^\tau(\cdot \mid x, t) = \mathrm{softmax}(z_T(x)_t/\tau)\,, \qquad q_S^\tau(\cdot \mid x, t) = \mathrm{softmax}(z_S(x)_t/\tau)\,.$$

The implemented output-distillation loss is

$$\mathcal{L}_{\mathrm{KD}} = \tau^2 \frac{\sum_{x,t} m_{xt} D_{\mathrm{KL}}(p_T^\tau(\cdot \mid x, t) \,\|\, q_S^\tau(\cdot \mid x, t))}{\sum_{x,t} m_{xt}}, \tag{14}$$

where m_{xt} masks padding. Thus the executable direction is teacher to student, $D_{\mathrm{KL}}(p_T^\tau \| q_S^\tau)$. All distillation runs use $\tau = 2$, multiply by τ^2, clip the optimized gradient norm at 1, and use AdamW.

Table 8 consolidates each family's models, data, schedule, seeds, and quality rule. Within a paired recorded-target intervention, the recorded-target arm and its matched permuted-target arm share initialization, data, seed, optimizer schedule, and final-step evaluation; comparisons with ordinary KD are claim-bearing only when language quality is also comparable. The sentence recorded-target runs retained no intermediate checkpoint frontier, and the headroom screen deliberately differs in initialization and step count, so neither comparison should be read as an objective-only effect.

Quality is measured only within a common tokenizer and scoring implementation. The predicted-target family uses a relative-perplexity criterion and the saved-student transfer assay adds a within-seed bits-per-byte threshold; the naturalistic full-parameter probe uses a forgetting guard and remains a three-participant mechanism study. The exact criteria and the rejected aggressive naturalistic schedule are recorded in Table 8.

C.2 Recorded-response intervention

For a recorded target y_i, the common forward path is

$$x_i \longrightarrow h_k(x_i) \longrightarrow W h_k(x_i) \longrightarrow \mathcal{L}_{\mathrm{target}},$$

where h_k is the attention-mask mean of the selected hidden state and W is a temporary linear map unless the loss is readout-free. The optimized objective is

$$\mathcal{L} = \lambda_{\text{text}}\mathcal{L}_{\text{text}} + \lambda_{\text{target}}\mathcal{L}_{\text{target}}, \tag{15}$$

with $\lambda_{\text{text}} = 1$. The unanchored sentence screen uses masked next-token cross-entropy for $\mathcal{L}_{\text{text}}$; the teacher-anchored sentence, capacity, contrastive, and naturalistic variants use Equation 14 when their Qwen teacher is supplied. The primary sentence intervention attaches its target branch at layer 12; Table 8 gives its weight and schedule.

For $\widehat{Y}, Y \in \mathbb{R}^{B\times D}$, the implemented squared-error target loss is

$$\text{MSE}(\widehat{Y}, Y) = \frac{1}{BD}\sum_{b=1}^{B}\sum_{j=1}^{D}(\widehat{Y}_{bj} - Y_{bj})^2.$$

The mean is over both minibatch rows and target coordinates, so λ_{target} is interpreted under this normalization.

In the primary LoRA regime, rank-16 adapters with scaling 32 and dropout 0.05 are placed in the model-appropriate attention and feed-forward projections. The pretrained base, input embeddings, and tied output matrix are frozen; the temporary readout and all adapters are optimizer parameters. The target loss attached at h_k reaches the readout and trainable adapters that produce h_k. It supplies no gradient to adapters in blocks after k or to the output head through this branch. Those later adapters still receive the text-loss gradient. After training, adapters are merged into an ordinary causal language model and the temporary readout is discarded. Evaluation freezes this merged model and fits a fresh ridge readout inside the held-out scoring folds.

The tested departures from that common regime are exact and narrow:

The exploratory sentence-loss screen also replaces only $\mathcal{L}_{\text{target}}$: cosine and squared-Pearson use a co-trained W, whereas linear CKA compares the batch geometry of h_k and the target without a readout. They were screening variants, not universal tests of geometry-based training. None displaces the primary MSE definition.

C.3 Synthetic-response intervention

The predicted-target run uses the same two-loss form as Equation 15, but its parameter path is materially different. For GPT-2 input x, the forward computation is

$$x \longrightarrow e(x) \longrightarrow h_6(x) \longrightarrow h_{7:12}(x) \longrightarrow q_S^{\tau}(\cdot \mid x),$$

with a temporary branch

$$\text{meanpool}(h_6(x)) \longrightarrow W_{\text{tmp}}\,\text{meanpool}(h_6(x)) \longrightarrow \mathcal{L}_{\text{target}}.$$

The teacher is frozen, but every student parameter and W_{tmp} is optimized. GPT-2 ties its input embedding to its language-model output matrix, so that shared retained matrix, all early blocks, all later blocks, and the output path can move through $\mathcal{L}_{\text{KD}}$. The target loss reaches W_{tmp}, the tied embedding, and blocks that produce h_6. The target branch does not traverse blocks 7–12 or the output-projection operation. Blocks 7–12 move only through output KD, whereas the tied embedding/output matrix also receives target-loss gradient through its input-embedding role. This distinction prevents the target branch from being misdescribed as back-propagating through the later student blocks.

Each target coordinate is standardized once from the 95,999 training rows before optimization. The aligned arm uses all 20,484 standardized TRIBE coordinates; no coordinate selection or PCA occurs in the training loss. The linear head maps the 768-dimensional pooled layer-6

Table 7: Recorded-target intervention paths. Schedules and language-quality rules are consolidated in Table 8. "Retained" names the component available to the later encoding assay, not evidence that the variant succeeded.

Variant	Target and gradient path	Retained object and scope
Co-trained mean squared error (MSE)	A temporary W is initialized per run and trained jointly in $\mathrm{MSE}(Wh_k, Y)$; the loss reaches W and the adapters that produce h_k. The recorded-target and matched permuted-target arms use the same path.	Merged student only; the five-ROI readout is discarded. This is the primary recorded-target form.
Frozen readout	The same mean MSE, but W_0 is ridge-fitted on untuned outer-training features with $\alpha = 100$ and then fixed; target gradients must therefore move the producing adapters.	Merged student only. Its recorded-target-minus-matched-permuted-target result is non-identifying and does not replace the MSE regime.
Contrastive	Symmetric InfoNCE compares normalized Wh_k with the paired five-ROI target using in-batch negatives; the temporary W and producing adapters are trained jointly.	Merged student only. The null is scoped to this low-dimensional target and is not a voxelwise contrastive test.
Adapter capacity	The same co-trained MSE path as the primary participant run, but with higher-rank adapters.	Merged student only. The manipulation did not move farther, so this is a capacity repetition, not evidence against every stronger intervention.
Naturalistic voxelwise	A participant-specific head maps pooled 25-word segments to independently selected reliable voxels under a 4-second response delay; its MSE reaches that head and the producing adapters.	Merged student only. Cells in which recorded-target tuning failed to improve over the untuned base, and also failed to beat the matched permuted-target arm, are lever failures rather than voxel or participant replications.
Full fine-tuning	The naturalistic voxelwise MSE and output-KD anchor jointly update every student parameter except the tied input/output embedding; no LoRA is used.	The full student is used immediately for held-out encoding evaluation and then released; the temporary voxel head is discarded, and only the result artifact is retained. The conditionally planned generic-text arm was not run because the manipulation check did not justify it.

state directly to this target. The aligned, block-permuted, and text-feature target arms all use the mean MSE defined above for $\mathcal{L}_{\text{target}}$. The ordinary-KD arm omits the temporary head. At the end of a target run, the head is intentionally discarded and only the causal language model is optionally saved. Held-out target recovery then fits a fresh standardized ridge readout from the frozen post-training student, in memory during the original run or from a saved checkpoint in later assays; Tuckute biological assays separately fit fresh nuisance-controlled readouts. Consequently, low training-head loss cannot enter either reported endpoint.

Four controls delimit what that intervention can show. First, every target arm has a seed-matched ordinary-KD arm initialized from the same pretrained student. The corresponding TRIBE and text-feature ordinary-KD artifacts were verified to be byte-identical before shared-baseline comparisons. Second, the aligned TRIBE arm has a saved ten-block target-permutation arm. That implementation can leave fixed rows and, for unequal blocks, duplicate one row while omitting another; it is retained as a defective sensitivity, not an exact-null trained control. Third, the nonbrain target mean-pools frozen `gpt2-medium` states and applies a seeded Gaussian projection to 20,484 coordinates. It matches dimension, corpus, model lineage, target-loss form, and schedule, but measured audits show mismatches in covariance geometry, baseline headroom, nuisance predictability, parameter displacement, and retained-representation movement. Initial-loss and gradient-scale comparability is unknown because those quantities were not retained. It therefore cannot support attribution to neural target content. Fourth, the exact-substrate diagnostic constructs a hash-fixed Sattolo row twin with no fixed points and matched within-fold row multisets. That twin is used only in the later exact-target measurement; no student was trained against it. The predicted-target family therefore has no valid shape- and optimization-matched trained nonbrain control.

C.4 Reproducibility

Table 8 collects the load-bearing training grid. It complements, rather than duplicates, the verified path–hash ledger in Table 6. The owning E record remains authoritative for commands, corrections, and whether an output is claim-bearing.

Table 8: Minimum training information needed to replay the load-bearing families, including their language-quality rules. Model names are the recorded Hugging Face identifiers.

Family	Models, data, and objective	Schedule and quality rule	Replay and retained evidence
Ordinary KD headroom	`gpt2-medium`→`gpt2`; 96,000/2,000 WikiText sentences; $\tau = 2$, $\lambda_{\text{KD}} = 1$.	Seeds 0–2; warm 1 epoch at 5×10^{-5}, cold 2 epochs at 3×10^{-4}; batch 16, length 64. Held-out perplexity is a convergence guard; warm and cold arms are not initialization- or step-matched.	`scripts/run_kd_alignment.py`; cold, warm, perplexity, and dissociation bundles in Table 6.
Sentence recorded target	Qwen2.5-0.5B student; five outer Tuckute folds; rank-16 LoRA; masked next-token cross-entropy when unanchored or output KD from Qwen2.5-1.5B when anchored, plus layer-12 target loss.	Seeds 0–2; 3 epochs, 2×10^{-4}, batch 16, primary $\lambda_{\text{target}} = 10$; five permutation draws in the participant assay. A fixed WikiText probe is reported, but no checkpoint quality frontier was retained.	`scripts/run_brain_lever.py`; primary and participant shards in Table 6.
Capacity and loss probes	Same Qwen sentence lineage; rank-64 MSE capacity probe or rank-16 symmetric InfoNCE.	Capacity: seeds 0–1, 6 epochs. Contrastive: seeds 0–1, 3 epochs, target weight 1, temperature 0.07. The same fixed WikiText quality probe applies.	`scripts/run_brain_lever.py` and `scripts/brain_loss.py`; capacity and contrastive shards in Table 6.
Naturalistic recorded target	Qwen2.5-0.5B; 14 tuning and 6 evaluation stories; 25-word segments; voxelwise MSE. The unanchored probe omits a teacher; the anchored sweep and full-parameter probe use `Qwen/Qwen2.5-1.5B`.	Unanchored: UTS01–03, seeds 0–1, two permutations, weight 10. Anchored sweep: UTS03, reliability > 0.7, seed 0, one permutation, three folds, weights $1, 3, 6$. Full-parameter gentle grid: UTS01–03, seeds 0–2, 2 epochs, 10^{-5}, batch 16, weight 10; the 3-epoch 5×10^{-5} schedule fails the forgetting guard.	`scripts/run_lebel_tune.py`; naturalistic probes and gentle grid in Table 6. The generic-text arm was conditional on a passed manipulation and was not run.
Predicted target	`gpt2-medium`→`gpt2`; 95,999/1,999 WikiText sentences; full student; layer-6 head to 20,484 standardized coordinates; target weight 0.1.	Seeds 0–5; 1 epoch, 24,000 steps, 5×10^{-5}, batch 4, length 64. Each target arm must remain within 5% relative perplexity of matched KD; saved-student transfer also requires $\lvert\Delta\text{bpb}\rvert \leq 0.005$ within seed.	`scripts/run_tribe_phase3.py`; target, text-feature, comparison, saved-student, and audit bundles in Table 6.

The repository replay environment requires the Python version declared in `pyproject.toml` and is resolved through the exact package versions and CUDA build recorded in `uv.lock`. Those files are the current replay pins, not a claim that every historical run serialized a complete as-run environment manifest. The result JSONs record model identifiers and training configurations, and later participant-transfer and exact-substrate bundles bind selected saved weights, inputs, runners, and analyses by hash. Most early runs do not record immutable upstream Hugging Face revision identifiers, so exact replay additionally depends on the retained cache or on the recorded saved-student hashes. This is a provenance limitation, not a reason to infer an unrecorded revision.

Only artifacts that carry a current manuscript claim, have an exact retained path, and have a path-bound SHA-256 appear in Table 6. Smoke runs, logs, temporary target caches, throughput probes, model-loading checks, incomplete control grids, and prospective validators remain diagnostic. The participant-transfer record contains result hashes without complete filename bindings, and the exact-substrate record binds several additional bundles without exposing every path–hash pair in the manuscript ledger; Appendix A states those omissions explicitly. The absence of a table row therefore means "not a verified manuscript-cited path," not "the experiment had no intermediate artifact."

D Estimators, Inference, and Formal Scope

This appendix separates three questions that are easy to conflate: what each encoding score predicts, which observations support its uncertainty statement, and which formal analogies apply. The empirical quantities are cross-validated predictive contrasts. They are neither causal effects nor direct estimates of information-theoretic objects.

D.1 Encoding estimators

Section 3.2 defines ordinary held-out R^2 and the cross-validated semipartial increment in Equations 1 and 2. The numerator of Equation 1 is the held-out sum of squared errors, $\mathrm{SSE}_j(M)$, and its denominator is the held-out total sum of squares, SST_j. Equation 2 measures the change in held-out linear predictive value when X is added to the specified nuisance design. Because the nuisance-only and full ridge estimators are separately regularized and evaluated out of sample, the difference can be negative. It is not a nonnegative variance decomposition.

For comparison, let $\mathcal{R}^2_{N,j}$ and $\mathcal{R}^2_{F,j}$ denote the population coefficients of determination for the nested, unregularized linear projections on N and $[N, X]$. The corresponding partial coefficient is

$$R^2_{\text{partial},j} = \frac{\mathcal{R}^2_{F,j} - \mathcal{R}^2_{N,j}}{1 - \mathcal{R}^2_{N,j}} = 1 - \frac{\mathrm{Var}(Y_j - \Pi_{[N,X]} Y_j)}{\mathrm{Var}(Y_j - \Pi_N Y_j)}, \tag{16}$$

where Π denotes the corresponding population linear projection and $\mathcal{R}^2_{N,j} < 1$. In a common-sample least-squares fit, the variance ratio is replaced by $\mathrm{SSE}_{F,j}/\mathrm{SSE}_{N,j}$. Equation 16 asks what fraction of the variance left by N is removed by X; Equation 2 asks how much total outcome variance is gained. The thesis reports Equation 2, not a transformed cross-validated partial coefficient.

Most recorded-response assays first compute Equation 1 per ROI or voxel, subtract reduced from full within that coordinate and fold, and then average the declared coordinates and folds equally. The continuity target-recovery assay also averages coordinatewise target R^2. The exact-substrate target-recovery assay is the exception: it sums SSE_j and SST_j across all 20,484 fixed target coordinates before taking $1 - \mathrm{SSE}/\mathrm{SST}$. That estimator weights coordinates by held-out variance and prevents near-constant coordinates from receiving the same weight as variable coordinates. Neither aggregation treats coordinates as independent repetitions.

Ridge regression supplies the fitted conditional mean. After train-fold centering and predictor standardization, suppose

$$y = X\beta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I), \qquad \beta \sim \mathcal{N}(0, \tau_\beta^2 I).$$

The posterior mode minimizes

$$\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2, \qquad \lambda = \sigma^2/\tau_\beta^2.$$

This is a maximum-a-posteriori estimator only under that Gaussian likelihood, an isotropic zero-mean coefficient prior, and the stated scaling convention. The implementations choose λ from a finite grid using training data, which is better understood operationally as predictive regularization than as a fixed scientific prior. The held-out R^2 then evaluates a predictive algorithm. Neither the Gaussian derivation nor a positive score makes the fitted coefficients causal, identifies a neural mechanism, or implies that model and brain implement the same computation.

D.2 Cross-validation and inference units

Every learned operation that can use stimulus-domain information is fitted on the relevant training complement. For each complete naturalistic story, fixed feature resampling and edge slicing, within-story feature and BOLD z-scoring, and finite-impulse-response feature-delay copies are constructed separately before cross-validation. Only train-complement PCA, assembled-design scaling, response centering, ridge selection, and readout fitting are learned for that assay. The sole declared exception is the training-target standardization reused as a frozen external object in the exact-substrate diagnostic; subsequent target PCA remains fold-local. Table 9 gives the load-bearing nesting order.

Table 9: Load-bearing cross-validation nesting and final inference units. Secondary layer, rank, and nuisance sensitivities are deferred to Appendices E–F and the owning records.

Assay	Readout nesting	Aggregation and inference
Baseline sentence measurement and headroom	Five contiguous 800/200 folds. Centering, scaling, PCA, and `RidgeCV` use the 800-sentence readout-training complement; one selected penalty is shared across outputs and applied to the 200-sentence test block.	Average declared ROIs and folds. The fixed sentence set supports an existence claim; stochastic headroom arms add technical-seed stability, not participant inference.
Recorded-target sentence intervention	Train the student on 800 outer-fold items and exclude the other 200 from model training. Within that 200-item block, fit transforms, select regularization, and evaluate a fresh readout over five contiguous inner subfolds.	Averaged-target claims use the five outer-fold means after seed aggregation. Participant variants first aggregate seed–fold cells within person and report the shared-fold axis separately.
Naturalistic story assays	Construct fixed feature resampling and edge slicing, within-story feature and BOLD z-scoring, and finite-impulse-response feature-delay copies separately for each complete story before cross-validation. On the remaining readout-training stories, learn only PCA, assembled-design scaling, response centering, one shared multi-output ridge penalty, and the readout. Reliability selection uses a separate repeated story.	Story blocks and voxels support within-person robustness only. The three-person intervention is a descriptive mechanism probe, not population inference.
Predicted-target uptake	Train on 95,999 WikiText sentences; select the fresh target-readout penalty on training data and evaluate once on 1,999 fixed held-out sentences.	Compare target-trained and ordinary-KD arms within matched seed, then summarize six technical seeds; coordinates are not repetitions.
Saved-student biological transfer	No retraining. In each of five contiguous Tuckute 800/200 folds, fit all transforms and fresh response readouts on the 800-item complement and apply them to the 200-item test block.	Aggregate six seeds and five folds within each of nine participants; participants are the biological units.
Exact-substrate diagnostic	Five outer folds reuse one train-only target-PCA basis for aligned rows and the frozen Sattolo row twin. Each multi-output ridge readout selects one scalar penalty on its own training complement; the test block is then touched once, and both paths in the composed student-to-target-to-response readout remain train-only.	Target-to-response and composed student-to-target-to-response assays aggregate to nine participants. Target recovery aggregates total SSE/SST and uses six technical seeds.

Regularization selection is nested inside each fitting set. The baseline sentence and naturalistic story assays run `RidgeCV` on the relevant complement for readout training and share one selected penalty across outputs. Recorded-target fresh readouts repeat that selection inside each inner training subfold. The exact-substrate diagnostic likewise forbids per-coordinate penalty selection: each aligned or twin multi-output readout selects one scalar penalty on its own training complement, while the paired paths reuse the train-fold target-PCA basis. Candidate layer grids, nuisance tiers, PCA ranks, reliability thresholds, participants, and ROIs are declared before scoring. The baseline screen reports best layers descriptively within its fixed grid; later recorded-response assays prospectively reuse those screen choices. The exact-substrate target-retention and composed student-to-target-to-response assays fix the trained layer 6 as primary and layer 7 as a locked, non-rescuing sensitivity.

The nesting order determines what can generalize. Repeated seeds vary optimizer randomness under one fixed procedure. Stimulus folds vary held-out blocks but share observations and have overlapping training complements. Voxels share one brain, spatial dependence, preprocessing, and stimuli. Target coordinates share the same generated target and student. None can replace a participant when the claim concerns people. Conversely, participants viewed through one fixed 1,000-sentence set do not by themselves establish stimulus-population generalization. The participant recorded-target assay therefore reports both participant and shared-fold robustness; saved-student transfer and exact-substrate response assays aggregate folds and seeds before participant inference; the naturalistic measurement retracts its voxel-bootstrap interval and keeps its within-person point contrast and story-block signs.

D.3 Uncertainty, power, and decision thresholds

Uncertainty follows the estimand rather than the largest available count. Table 10 maps the load-bearing result families to their intervals, tests, and decision rules. Unless stated otherwise, intervals are two-sided 95% intervals.

Table 10: Inferential distinctions and decision rules for the load-bearing estimands. Secondary sensitivity inventories are reported in Appendices E–F and the owning records.

Estimand family	Unit and uncertainty	Multiplicity and decision rule
Controlled measurability	Sentence assays report fold dispersion and trained–untrained controls. The naturalistic assay retains a within-person gap and five story-block signs; its voxel-bootstrap interval is retracted.	Layer and nuisance grids are fixed; selected voxels and model–substrate estimates are not repetitions. The stage establishes assay-bounded existence, not population evidence.
Headroom and averaged-target pairing	Headroom uses a fold–seed bootstrap conditional on convergence and family-cluster bootstrap with Fisher intervals for cross-family quality. Averaged-target pairing averages seeds within outer fold before a five-fold t interval.	Flat seed–fold intervals are superseded. The 0.33/0.80 retention bands and planned +0.006 80%-power MDE diagnose the design, not the objective.
Participant recorded-target assays	Sentence effects are participant medians over seed–outer-fold cells, with a participant t interval and exact one-sided sign test; a shared-fold t interval and cluster bootstrap address stimulus dependence. Naturalistic seeds are technical repeats in three descriptive participant probes.	Participant and shared-fold readings must agree for a population claim; observed-variance MDEs bound only scoped nulls. Failed manipulation checks or quality criteria stop the branch.
Synthetic-target uptake and brain-response-specific attribution	Six matched-seed gains use descriptive intervals and declared exact magnitude-preserving sign flips. Comparator checks use fixed samples and frozen margins, without coordinate-level p values.	A 5% relative-perplexity criterion and later $\lvert\Delta\text{bpb}\rvert \leq 0.005$ threshold constrain quality. Uptake and movement provide no biological replication and cannot repair an invalid comparator.
Participant biological transfer	Nine participant effects receive a t interval, one-sided upper bound, exact sign test, and Wilcoxon test; seeds and blocks are descriptive.	The MDE and upper bound are compared with the inherited +0.002 unique-R^2 threshold, distinguishing power limitation from practical exclusion only for this cohort and assay.
Exact-substrate diagnostic	Target-to-response and composed student-to-target-to-response quantities use participant t intervals and exact signs. Target recovery uses descriptive seed intervals and an exact 2^6 sign flip for the paired target-trained-minus-KD contrast.	Bonferroni family-95% intervals and one-sided simultaneous bounds are separate for target measurability, target recovery, and student-to-target-to-response families. The +0.002 rule applies only to this fixed linear route.
External utility	Eight technical seeds support paired arm contrasts and an observed-variance 80%-power MDE for prerequisite representation change and out-of-domain perplexity ratios.	Controls are matched as declared; task families are not biological repetitions, and downstream nulls remain bounded to the tested corpora and student–teacher pair.

Power is always computed for the future contrast at its intended unit. The original naturalistic follow-up calculation used unmatched absolute fold variability and is retracted. The participant recorded-target and saved-student transfer assays instead use between-participant variability for participant claims; external utility uses seed variability for a fixed technical procedure. An MDE says what a design could reliably resolve under its variance model. It does not say that smaller effects are zero or that larger effects matter biologically.

The +0.002 rule illustrates this boundary. It was inherited prospectively to decide whether the participant-positive branch merited a new matched-control build, then reused as a frozen reference in the exact-substrate diagnostic. It supports statements such as "the simultaneous upper bound misses the continuation rule on this fixed assay." It does not support "effects below +0.002 are biologically irrelevant," conversion to another target scale, or retrospective selection of a friendlier nuisance model, layer, or subgroup.

D.4 What information theory does and does not imply

Conditional information and partial variance. For scalar Y, jointly Gaussian (X, Y, Z), correctly specified population-linear conditional means, and unregularized population residual variances,

$$I(X; Y \mid Z) = -\frac{1}{2}\log\bigl(1 - R^2_{\text{partial}}\bigr)\,, \qquad \Delta\mathcal{R}^2_{\text{semi}} = (1 - \mathcal{R}^2_N)R^2_{\text{partial}}. \tag{17}$$

The multivariate analogue replaces the scalar residual-variance ratio with a log-determinant ratio. These are distributional identities under the stated Gaussian model; the thesis instead estimates finite-sample, cross-validated semipartial differences from separately regularized ridge fits [28]. Those differences may be negative and depend on the nuisance design, rank, penalty, and held-out substrate; they are operational incremental-prediction estimates, not estimates or lower bounds of $I(X; Y \mid Z)$.

Data processing. The data-processing inequality would let a predicted target T upper-bound what a trained state H retains about a recorded response B only under $B \to T \to H$, equivalently $B \perp H \mid T$, which implies $I(B; H) \leq I(B; T)$. Direct teacher compression analogously requires $B \to H_T \to H_S$. Neither chain follows from the objective: TRIBE, the teacher, and the student share text-derived information, and the student has a direct stimulus/corpus path not mediated by the target or teacher state. Conditioning on the stimulus does not restore the chain when the target is a deterministic or nearly deterministic function of that stimulus. Data processing therefore motivates a preservation test only after conditional independence is defended; it does not predict the sign of a cross-validated encoding contrast [28].

Generalization and rate–distortion. Let Z^n be an i.i.d. sample and let a learning algorithm map it, possibly randomly, to parameters W. If the loss $\ell(w, Z)$ is σ-sub-Gaussian under the population distribution for every fixed w, and $L(W)$ and $\widehat{L}(W)$ denote population and empirical risk, then expectation over the sample and algorithm randomness gives

$$\Bigl|\mathbb{E}[L(W) - \widehat{L}(W)]\Bigr| \leq \sqrt{\frac{2\sigma^2 I(W; Z^n)}{n}}$$

This information-theoretic generalization bound [29] grows with algorithm–sample information, which this experiment does not estimate. The bound is therefore an organizing constraint, not an explanation of an observed result: it neither guarantees that a neural target improves generalization nor makes the benefit scale with a "brain-information budget."

Classical rate–distortion theory defines

$$R(D) = \min_{P_{\widehat{S}|S}:\, \mathbb{E}[d(S,\widehat{S})] \leq D} I(S; \widehat{S})$$

In this classical object [28], parameter count, optimizer steps, and perplexity are not the mutual-information rate, and the target loss is not established as a source-coding distortion. The manuscript therefore uses "rate–distortion-style" only for the empirical Pareto relation among model budget, language quality, and controlled predictivity. No theorem about $R(D)$ guarantees that neural supervision moves that frontier favorably.

Output matching and internal alignment. Equation 14 constrains the student's output distribution, not internal-state geometry, because many parameterizations realize the same next-token distribution:

$$D_{\text{KL}}(p_T^\tau \| q_S^\tau) \to 0 \;\not\Rightarrow\; H_S \approx H_T \;\not\Rightarrow\; \Delta R^2(H_S \to B) = \Delta R^2(H_T \to B).$$

The first non-implication is architectural; the second also depends on the evaluation target, nuisance model, and readout. Language quality must therefore be measured, but a matched output score cannot itself certify biological alignment.

Low auxiliary target loss is no stronger: a temporary head may absorb it, or a nuisance-predictable target may be learned without a participant-general incremental relation to recorded responses. Held-out target uptake and retained-student movement show learning and change, not neural-content attribution or biological transfer without a valid comparator and participant endpoint. In canonical stage order, exact-target practical linear measurability fails its continuation rule; target uptake and retained-student movement pass; brain-response-specific attribution remains unsupported because the comparator is non-identifying; and biological transfer is near zero under the tested assay.

E Supporting Analyses, Failed Instruments, and Future Designs

The experiments below answer narrower questions than the main result chain. Some validate an implementation, some test a proposed rescue mechanism, and some stop because a measurement or design prerequisite fails. A stopped route is not converted into a negative intervention result, and an unexecuted design contributes no empirical evidence.

E.1 Pipeline validation and apparatus checks

Synthetic end-to-end pipeline validation. E001 combined real Pereira sentences [26] with a synthetic response constructed from teacher features, named nuisance variables, and noise. It exercised the entire initial path: ordinary and target-guided KD, the temporary target head, middle-layer extraction, contiguous held-out splitting, fold-local ridge fitting, matched-rank static and contextual feature blocks, nuisance subtraction, and paired aggregation over multiple optimization seeds. The expected teacher–student ordering, small nuisance contribution, stable raw-versus-unique behavior, and deterioration under excessive target weight all behaved coherently.

Those observations validate plumbing rather than a scientific hypothesis. The response was synthetic and partly generated from the same teacher features that ordinary KD already encourages the student to preserve. The run therefore cannot show that an LM predicts recorded activity, that the chosen nuisances suffice on real stimuli, or that a target loss improves a student. Its lasting contribution is narrower: it exposed a duplicate configuration key, verified that the evaluation tail was excluded from target-guided training, and showed that the analysis did not automatically reward the target arm. E002, which substituted recorded responses and trained–untrained controls, begins the scientific evidence.

E.2 Response averaging and intervention robustness

Response averaging changes both reliability and the estimand. E010 initially varied the number of participants contributing to a nested average and appeared to show a rising intervention contrast. Because every larger average contained the smaller one, participant identity, response reliability, and target construction changed together. The corrected random-subset analysis within E010 replaced that sequence with random participant subsets; their overlapping distributions did not preserve a stable monotone relation between participant count and the intervention contrast. The nested curve therefore does not identify a causal averaging dose.

E014 held the frozen representation and encoding procedure fixed while averaging increasing numbers of response targets. Controlled predictivity rose, as expected when idiosyncratic and measurement noise is suppressed and the shared stimulus-evoked component is emphasized. This is a legitimate signal-to-noise benefit for a group-response estimand, not evidence that averaging manufactures a false score. It also cannot supply participant replication or rescue the individual-response intervention tested by E008.

The tested intervention rescues do not overturn the participant result. E005 varied auxiliary-loss strength against the participant-averaged target. Near-quality-matched settings retained only a fold-sensitive trend; the setting with the clearest fold-level increase materially degraded language quality. It therefore describes a quality–alignment trade-off on a group-average endpoint, not a participant-general gain.

E011 tested the insufficient-adapter-capacity objection by increasing LoRA rank and training duration. The effective representation movement did not become stronger, and the participant estimate remained centered near zero; this closes the tested capacity repetition, not every higher-capacity intervention. E013 changed the loss geometry to a symmetric contrastive objective at the five-ROI target and separately tried a naturalistic voxelwise target. The contrastive participant estimate remained null, while the voxelwise path failed its real-over-base manipulation check and its comparison with the matched permuted-target arm, so the latter is a lever failure rather than a biological null. E017 then replaced adapters with full-parameter tuning: an aggressive schedule caused forgetting, and a gentler quality-preserving three-participant probe did not establish a target-specific gain. Together these experiments weaken the specific rescue hypotheses tied to the tested adapter-capacity setting and loss form, the attempted voxelwise induction route, and adapter-only parameterization. They do not show that all stronger objectives fail, and none supplies a valid positive participant-level contrast that would change E008.

E.3 External and cross-modal probes

Local Negi-style intervention diagnostic. E019 implemented a faithful-as-feasible version of the differentiable temporal prediction head and target-specificity logic used by Negi et al. [9]. On the available LeBel data [24] and decoder student, gentle tuning produced no stable real-target advantage, the intended quality control was not in fact quality matched, and stronger schedules degraded encoding or language quality. The result is useful corroboration that failure was not confined to the thesis's original five-ROI MSE objective.

It is not a reproduction or refutation of the source study. The local test changes the participant pool, stimulus and response substrate, encoder-versus-decoder architecture, language setting, training scale, and parts of the temporal implementation. It also never obtained the positive local lever needed for the planned matched-quality analysis. The supported conclusion is only that this Negi-style route did not provide a target-specific rescue under the tested local controls.

Reduced Moussa-style speech pilot. E022 started the pretrained and brain-tuned arms from the same Wav2Vec2-base checkpoint and asked whether brain tuning first produced a reliable phoneme-classification gain before constructing a matched-quality shrink test. The reduced, single-participant LeBel setup did not cross its prospective phoneme-gain criterion, although the fMRI target loss fell, confirming that adaptation ran. The downstream matched-quality specificity test was therefore correctly left unbuilt: without a reproduced positive, there was no gain whose specificity could be tested.

The source study used a larger multi-participant speech protocol, more neural data, different timing and training details, and several speech-model families [8]. E022 is consequently a weak local non-reproduction, not evidence against the published result. E019 and E022 together show that the local apparatus did not furnish an external positive demonstration; they do not settle the broader literature.

E.4 Cognitive and privileged-target routes

The shared-response reference was not a usable ceiling. E020 asked whether a leave-one-participant-out response average could approximate $\mathbb{E}[Y \mid S]$, so that residual predictivity would isolate structure beyond the shared stimulus-predictable response. The reference was too unreliable, and temporal leakage and autocorrelation preserved apparent residual signal. With the cleanest gapped and semantically controlled construction, the trained–untrained residual advantage disappeared. This is a reference-reliability and attribution failure: it invalidates this empirical ceiling instrument but neither proves participant-specific residual structure nor establishes a universal ceiling. Adding more stories would not repair the small participant reference or identify the latent conditional mean, so the broader build was stopped.

Reading-time content was not separated from target shape. Raw and surprisal-residualized reading time initially appeared to yield similar frozen-probe learning advantages over simple shuffled or random controls. Those controls did not match the residual target's autocorrelation and marginal shape. The final phase-randomized control preserved those statistical properties while removing behavioral content and tied the reading-time residual. The supported result is therefore a generic signal-shape or regularization advantage, not a cognition-specific benefit. The sequence also replaced an early downstream metric that failed its positive control with a validated frozen probe, so the final failure is attributional rather than a claim that reading time contains no reliable signal.

The gaze-first route failed its preconditions. The planned LUPI intervention required a natural-reading gaze summary that was reliable, informative about relation labels, and evaluable with a leakage-free grouped split. Natural-reading gaze was reliable enough to measure but largely tracked sentence structure and remained uninformative about the label across linear, nonlinear, pooled-participant, and within-participant probes. Task-directed gaze was predictive, but that contrast was consistent with label or task-intent leakage rather than portable privileged information. The corpus also contained too few independent paragraph groups for a stable paragraph-disjoint learning curve. Because both signal informativeness and evaluation substrate failed before intervention, the multi-arm student build was correctly stopped and the untested EEG branch received no verdict.

E.5 Unexecuted future designs

Objective-specific shedding design. E023 defines the quantity that the headroom screen could not identify: a KD student's vertical residual below a same-lineage alignment–quality frontier, paired with ordinary fine-tuning from the same initialization. An analysis-only slope and knee screen found one lower-quality pilot region but did not establish the intended high-quality, locally flat regime. No paired training intervention was authorized. E023 therefore contributes a design for separating objective-specific loss from ordinary quality loss, not evidence that KD has such an effect.

Reserved matched auxiliary-target design. E027 was a reserved identifier for asking whether the neural proxy outperformed a geometry- and learnability-matched nonbrain target. It has no owning experiment record or authorized design. Had the branch activated, E026 implies that a valid comparator would also have needed to match optimization difficulty, initial gradients, and student movement without copying itemwise neural-proxy coordinates. E025 missed the participant-positive activation rule, and E026 showed why the saved text-feature target could not substitute for the missing control, so no E027 target was constructed or student trained.

Outcome-blind external-transfer designs. E028 would test whether supervision from real training fMRI outperforms a phase-randomized shape twin and a geometry-calibrated nonbrain target on an independently measured cross-modal endpoint, following the external transfer setting of Vaidya et al. [11]. Only protocol metadata and outcome-blind synthetic development tests were completed. The E028 development run opened no neural endpoint values, trained no target student, and scored no biological endpoint; the synthetic values validate plumbing only.

E029 would ask whether a prospectively computed directional certificate predicts the sign and ordering of auxiliary-target transfer before biological outcomes are opened. It remains a design draft that has not received independent pre-execution review and is conditional on an informative E028 regime. Its deterministic algebra validator is not a result. Reserved labels, outcome-blind development, and prospective designs matter for provenance and stopping discipline, but none enters the empirical verdict.

F Sensitivity Analyses and Full Numerical Tables

This appendix separates sensitivities that changed the interpretation from complete participant and saved-student seed values that permit aggregation checks. The tables do not create new inference units: participants support biological generalization, whereas blocks and optimization seeds diagnose stimulus and training-procedure stability. All row values come from the hash-bound retained artifacts listed in Appendix A.

F.1 Sensitivity hierarchy

Table 11 reports only axes capable of changing a main claim. The full layer screens, training grids, and diagnostic leaves remain in their owning E records.

Table 11: Claim-relevant sensitivity hierarchy. “Changed” means that the sensitivity narrowed or reversed an earlier interpretation; “confirmatory” means that it left the scoped main disposition unchanged.

Axis	What was varied	Effect on the claim
Representation rank	The headroom screen repeated its fixed-layer assay over the declared PCA ranks.	**Changed scope.** The ordering of the model arms survived, but absolute magnitudes and some signs did not; headroom is rank-conditional and cannot be quoted as an invariant amount.
Auxiliary weight and language quality	E005 paired each target-loss weight with its own permuted-target arm and fixed quality probe.	**Changed scope.** Near-quality-matched weights remained fold-uncertain; the clearest fold-positive setting paid a substantial language-quality cost. The result is a trade-off on an averaged target, not a free frontier shift.
Quality band and model family	E015 compared the full-range association, −0.783, with the capable-band estimate, −0.501, whose interval is [−0.859, +0.188].	**Changed scope.** Removing near-floor models attenuated the association and made the capable-band estimate uncertain. Language quality remains a mandatory control, not a deterministic explanation within the operative band.
Layer, nuisance set, and aggregation	E025 locked a later biological layer as primary and the trained layer as non-rescuing; it added imageability and a broader contextual stress set. E030 crossed its primary and adjacent layers with base, imageability-complete, and broader nuisance sets, and compared variance-weighted with equal-coordinate target aggregation.	**Confirmatory.** No E025 variant produced a participant rescue. E030 retained positive absolute target extractability, unresolved incremental retention and composed student-to-target-to-response quantities, and failure of the exact-target practical measurability rule; alternative nuisance and aggregation choices had no rescue authority.
Participant and subgroup	E008 replaced a response average with participant-first inference. E025 retained the predeclared training and held-out subgroups and removed the highest-baseline participant as a sensitivity. E030 reported every participant and leave-one-participant-out mean.	**Changed the main claim.** E008 removed support for a participant-general recorded-target gain. In E025, dropping the influential participant gives −0.000131; in E030, every leave-one-participant-out aligned-above-nuisance mean remains negative.

Continued on next page

Table 11 (continued)

Axis	What was varied	Effect on the claim
Held-out block	E006 reported story-block signs within one participant; E008, E025, and E030 reported shared sentence-block values and leave-one-block-out means.	**Changed or bounded scope.** E008's block-axis mean is $+0.00026$ with interval $[-0.0004, +0.0009]$, and no leave-one-block result establishes a positive effect. E025 is not driven by one block, but still fails across participants. E030's measurability failure is stable to every block deletion.
Utility-mediator seed	E009 repeated the target arm versus matched permuted-target arm representation contrast over $n = 8$ technical seeds.	**Changed scope.** 5/8 signs were positive. Seed 0 contributed $+0.063$, whereas the other seven averaged approximately $+0.004$. The utility analysis therefore tests whether value survives a weak, seed-unstable mediator, not whether a reliably induced target-specific change improves utility.
Saved-student optimization seed	Recorded-target and saved-student analyses repeated the fixed procedures over technical seeds; target uptake, movement, transfer, retention, and composed student-to-target-to-response contrasts were kept separate.	**Confirmatory only.** Seed agreement can establish reproducible optimization or manipulation, but it cannot substitute for participant agreement. Complete E025/E030 saved-student seed values appear in Table 14.
Checkpoint and control construction	Sentence interventions retained final-step students rather than an intermediate quality frontier. The saved predicted-target block permutation and text-feature comparator were audited after training.	**Limitation and correction.** No post hoc checkpoint can identify an objective-only frontier. The block permutation is defective, and the text-feature target fails measured-axis comparability, so neither can carry a neural-content attribution claim.

Representation rank, auxiliary weight and language quality, quality band and model family, participant and subgroup, held-out block, and E009 utility-mediator seed heterogeneity changed or bounded what could be claimed. The locked E025/E030 layer, nuisance, and aggregation grids and the saved-student seed checks are confirmatory: they show that the scoped disposition is not an artifact of one nearby analysis or unstable optimization. The checkpoint and control-construction row records a limitation and a correction. None authorizes replacing a prospective primary estimator with a more favorable variant.

F.2 Participant-level values

For E008, let d_{uks} be the real-target score minus the mean of its matched permuted draws for participant u, held-out block k, and seed s. The participant effect $e_u = \text{median}_{k,s}\, d_{uks}$ is the median over 15 paired seed-by-outer-fold cells, and the cohort estimate is the mean of the 9 participant medians.

For E025, let T, X, and K denote the TRIBE-target, text-feature-target, and ordinary-KD scores. Each displayed participant tuple is ordered as $(T - X,\ T - K,\ X - K)$ after averaging seeds and outer folds within participant. At artifact precision, the identity $T - X = (T - K) - (X - K)$ makes clear when a relative contrast is positive because the comparator moved downward; the displayed entries are rounded independently.

Table 12: Complete E008 and E025 participant values in unique-R^2 units. E008 and E025 have different targets and within-participant aggregators and are not pooled.

UID	E008 e_u	E025 $(T-X,\ T-K,\ X-K)$
797	+0.00002885	−0.00023801, −0.00005311, +0.00018490
837	+0.00100412	+0.00227168, +0.00226591, −0.00000577
841	−0.00018259	+0.00000452, −0.00078813, −0.00079265
848	−0.00063374	−0.00002346, −0.00002667, −0.00000320
856	−0.00037809	−0.00022167, −0.00036255, −0.00014087
865	−0.00058994	−0.00064994, −0.00108762, −0.00043767
875	+0.00070575	+0.00009116, −0.00006339, −0.00015455
876	+0.00011067	−0.00013023, −0.00014342, −0.00001319
880	+0.00083826	+0.00012305, +0.00013517, +0.00001212
Mean	+0.00010	(+0.000136, −0.000014, −0.000150)

The E008 signs are mixed around a small cohort mean. E025 is also heterogeneous: the relative mean is small and positive, but the direct TRIBE-minus-KD mean is −0.000014, and removal of the most influential participant reverses the relative point estimate. The table therefore supports the participant-level conclusion rather than a claim that every participant has exactly zero effect.

E030 uses the same participant identifiers but answers different mechanism-localization questions. Under the primary imageability-complete analysis, each row reports five quantities in order: exact-target unique signal above nuisance; aligned-minus-frozen-twin specificity; absolute composed student-to-target-to-response overlap; the TRIBE-minus-KD student-to-target-to-response overlap increment; and the aligned-minus-frozen-twin overlap increment. The first two quantities average outer folds within participant, while the three composed student-to-target-to-response quantities average seeds and outer folds within participant.

Table 13: Complete E030 participant values in unique-R^2 units, ordered as exact-target unique signal, aligned-minus-frozen-twin specificity, absolute composed student-to-target-to-response overlap, TRIBE-minus-KD overlap increment, and aligned-minus-frozen-twin overlap increment.

UID	E030 quantities in caption order
797	−0.00266759, −0.00031777, +0.00045651, +0.00018799, +0.00019106
837	−0.00448823, −0.00785204, +0.00484230, +0.00029143, +0.00026774
841	−0.02042083, +0.00588154, +0.00038073, +0.00004280, +0.00003954
848	−0.00001541, +0.00158126, +0.00010189, −0.00005708, −0.00005644
856	−0.01394534, −0.00066703, −0.00040275, −0.00005387, −0.00005261
865	+0.00208088, +0.00684375, +0.00216009, −0.00006632, −0.00005151
875	+0.00266981, +0.01519279, +0.00447180, −0.00018830, −0.00019123
876	−0.00660269, −0.00590127, −0.00126093, +0.00008691, +0.00008438
880	+0.00293994, +0.00106256, +0.00043030, +0.00003540, +0.00003336
Mean	(−0.004494, +0.001758, +0.001242, +0.000031, +0.000029)

The participant rows show why the two exact-target conclusions differ. The exact-target unique signal misses its frozen practical rule and is negative under every participant deletion, whereas aligned-minus-frozen-twin specificity contains substantial effects of both signs and remains unresolved. Absolute composed student-to-target-to-response overlap also remains unresolved. The TRIBE-minus-KD and aligned-minus-frozen-twin overlap increments are much smaller and mixed; neither repairs the earlier exact-target measurability failure or identifies its cause.

F.3 Technical-seed values

E025 seed rows report, in order, TRIBE minus text-feature target, TRIBE minus ordinary KD, and text-feature target minus ordinary KD; each averages participants and outer folds within a saved seed. E030 seed rows report absolute target extractability, incremental TRIBE-minus-KD retention, the TRIBE-minus-KD student-to-target-to-response overlap increment, and the aligned-minus-frozen-twin overlap increment. Target extractability and incremental retention aggregate held-out target recovery within each seed, while the two composed student-to-target-to-response overlap increments average participants and outer folds within each seed.

Table 14: Complete technical-seed values for the E025 and E030 endpoints only. E025 order: TRIBE minus text-feature target, TRIBE minus ordinary KD, and text-feature target minus ordinary KD. E030 order: absolute target extractability, incremental TRIBE-minus-KD retention, TRIBE-minus-KD student-to-target-to-response overlap increment, and aligned-minus-frozen-twin overlap increment. Seeds diagnose fixed training procedures, not biological or model-population replicates.

Seed	E025 quantities in caption order	E030 quantities in caption order
0	−0.00015727, −0.00035769, −0.00020042	+0.03965524, +0.00013878, −0.00003781, −0.00004017
1	+0.00032614, +0.00003556, −0.00029058	+0.04096723, +0.00103098, −0.00004152, −0.00004668
2	+0.00004025, +0.00000892, −0.00003133	+0.03932328, −0.00159873, −0.00001666, −0.00001488
3	+0.00025080, +0.00015307, −0.00009773	+0.04291468, −0.00234707, +0.00006481, +0.00006414
4	+0.00018344, −0.00006376, −0.00024721	+0.03920675, −0.00038085, +0.00014778, +0.00014581
5	+0.00017470, +0.00014137, −0.00003333	+0.04228561, −0.00197014, +0.00006938, +0.00006797
Mean	(+0.000136, −0.000014, −0.000150)	(+0.040725, −0.000855, +0.000031, +0.000029)

Before the E025 participant endpoint was interpreted, all target-learning and retained-student movement checks passed. The largest saved-arm language-quality difference, 0.002579, remained within the frozen 0.005 margin. The manipulation therefore succeeded even though the direct biological contrast did not. In E030, absolute target extractability is consistently positive, whereas incremental retention and both student-to-target-to-response overlap increments have mixed signs and intervals containing zero. This separates absolute target extractability from incremental retention or transfer along the composed student-to-target-to-response route.

E008 has no corresponding seed table because its primary participant statistic is a median over the crossed seed–block cells; a seed-only vector does not reconstruct that nonlinear participant aggregate. Its seed variability remains part of each participant value, while the separately reported shared-block summary tests the other crossed axis. This omission prevents a technical summary from being mistaken for another biological analysis.

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